

Statistical Methods in Imaging (SMI) 2026 – Program

PROGRAM AT A GLANCE

Venue: [Michigan League](#) Room Assignments: *Track A:* Michigan room 2nd floor; *Track B:* Hussey room 2nd floor; *Track C:* Room 4, 1st floor.

Time	Day 1 (June 1, 2026)	Day 2 (June 2, 2026)	Day 3 (June 3, 2026)
7:30–8:00 AM	Welcome, registration & coffee	Coffee	Coffee
8:00–9:20 AM	Track A · S1 Scalable, Interpretable, and Reproducible Methods for Neuroimaging Analysis Organizer: Jian Kang · Chair: Jian Kang	Track A · S12 Methods for Statistical Imaging Data with Applications to Physical Sciences Organizer: Yang Chen · Chair: Yang Chen	Track A · S24 Network methods in neuroimaging Organizer: Liza Levina · Chair: Liza Levina
	Track B · S2 Advances in methods for cancer imaging: From single-cell data to sustainable AI Organizer: Andrew Whiteman · Chair: Tim Johnson	Track B · S13 From Connections to Understanding: Brain Network Models in Neuroscience Organizer: Heather Shappell · Chair: Heather Shappell	Track B · S25 AI for Medical Image Analysis III Organizer: Kayvan Najarian · Chair: Ivo Dinov
	Track A · S3 AI for Medical Image Analysis I Organizer: Kayvan Najarian · Chair: Kayvan Najarian	Track A · S15 From Pixels to Physics: Statistical Methods for Astronomical Images Organizer: Jeffrey Regier · Chair: Jeffrey Regier	Student Paper Session Ballroom
9:30–10:50 AM	Track B · S4 Statistical Innovation in Multiplex Imaging: Methods for Robust Discovery Organizer: Yue Wang · Chair: Yue Wang	Track B · S16 Statistical and AI Methods for Heterogeneous and Irregular Imaging Data	

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	Track C · S5 Alignment, Calibration, and Robust Inference in Neuroimaging Organizer: Panpan Zhang · Chair: Panpan Zhang	Organizer: Chunming Zhang · Chair: Chunming Zhang	
		Track C · S17 Achieving Reliable Inference and Prediction from Noisy Imaging Data through Structure Learning Organizer: Shuo Chen · Chair: Sharmistha Guha	
11:00–11:15 AM	Break & Welcome Address	Break	Break
11:15 AM–12:15 PM	Keynote 1 (11:30 AM): Prof. Hernando Ombao (Michigan League Ballroom)	Keynote 2: Prof. Nicole Lazar (Michigan League Ballroom)	Keynote 3: Prof. Douglas Noll (Michigan League Ballroom)
12:15–1:30 PM	Lunch (12:30 PM)	Lunch	
	Track A · S6 Spatiotemporal Models for Brain Structure, Function, and Connectivity Organizer: Mark Fiecas · Chair: Tim Johnson	Track A · S18 Recent Advances in Complex Modeling of Brain Imaging and Networks Organizer: Xi Luo · Chair: Jian Kang	
1:30–2:50 PM	Track B · S7 Advanced Statistical and Machine Learning based methods for Spatial imaging and omics Organizer: Satwik Acharyya · Chair: Andrew Whiteman	Track B · S19 Next-Generation Statistical and Inferential Methods for Neuroimaging and Multimodal Data Integration Organizer: Haochang Shou · Chair: Haochang Shou	
	Track C · S8 Principled Statistical and Machine Learning Methods for Brain Imaging Inference	Track C · S20 AI for Medical Image Analysis II Organizer: Kayvan Najarian · Chair: Shuo Chen	

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Organizer: Zhengwu Zhang · Chair:
Zhengwu Zhang

3:00–4:20 PM	Track A · S9 Advanced statistical and AI models for complex and graphical brain images Organizer: Yize Zhao · Chair: Yize Zhao	Track A · S21 Recent advancements in statistical methods for MRI Organizer: Benjamin Risk · Chair: Benjamin Risk
	Track B · S10 Emerging Statistical Techniques for Longitudinal and Multimodal Imaging Analysis Organizer: Sharmistha Guha · Chair: Ivo Dinov	Track B · S22 Statistical Network Analysis for Neuroimaging Organizer: Dan Kessler · Chair: Dan Kessler
	Track C · S11 Modern Statistical Methods for Neuroimaging Organizer: Xi Luo · Chair: Yi Zhao	Track C · S23 Recent developments on neuroimage analyses for use in digital twins Organizer: Ansu Chatterjee · Chair: Emily Hector
4:30–6:30 PM	Training Tutorials & Short Courses Empowering Large Language Models with Statistics: A Practical Tutorial for Medical Imaging Khashayar Namdar (University of Toronto & Dalhousie University, Canada), Pascal Tyrrell (University of Toronto) Room 4, 1 st floor Manifold learning and dimension reduction for imaging data Chunming Zhang (University of Wisconsin–Madison) Michigan Room, 2 nd floor Methods for FMRI processing, quality control and group analysis in AFNI	<u>SMI-2026 Posters and Mixer Event</u> Ballroom

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**Paul Taylor & Gang Chen (Scientific and
Statistical Computing Core, NIMH, NIH)**

Ballroom, 2nd floor

Introduction to P300 Brain-Computer

Interface Data Analysis Challenges

Jane Huggins (University of Michigan)

Hussey Room, 2nd floor

Statistical Methods in Imaging (SMI) 2026 – Program

Detailed Program

DAY 1: June 1, 2026

7:30–8:00 AM

Welcome, registration & coffee

8:00–9:20 AM

Track A · S1

Scalable, Interpretable, and Reproducible Methods for Neuroimaging Analysis

Organizer: Jian Kang · Chair: Jian Kang

Yuliang Xu (University of Chicago)

Bayesian Image Regression with Soft-thresholded Conditional Autoregressive Prior

In the analysis of brain functional MRI (fMRI) data using regression models, Bayesian methods are highly valued for their flexibility and ability to quantify uncertainty. However, these methods face computational challenges in high-dimensional settings typical of brain imaging, and the often pre-specified correlation structures may not accurately capture the true spatial relationships within the brain. To address these issues, we develop a general prior specifically designed for regression models with large-scale imaging data. We introduce the Soft-Thresholded Conditional AutoRegressive (ST-CAR) prior, which reduces instability to pre-fixed correlation structures and provides inclusion probabilities to account for the uncertainty in choosing active voxels in the brain. We apply the ST-CAR prior to scalar-on-image (SonI) and image-on-scalar (IonS) regression models—both critical in brain imaging studies—and develop efficient computational algorithms using variational inference (VI) and stochastic subsampling techniques. Simulation studies demonstrate that the ST-CAR prior outperforms existing methods in identifying active brain regions with complex correlation patterns, while our VI algorithms offer superior computational performance. We further validate our approach by applying the ST-CAR to working memory fMRI data from the Adolescent Brain Cognitive Development (ABCD) study, highlighting its effectiveness in practical brain imaging applications.

Luling Altan (Indiana University)

Alzheimer's Disease Subtype Identification from Spatial Tau-PET Heterogeneity Using Self-Supervised Deep Learning

The multifactorial and heterogeneous nature of Alzheimer's disease (AD) complicates efforts to modify disease progression. We introduced a self-supervised deep learning framework with autoencoder-based representation learning and iterative pseudo-labeling to capture the spatial heterogeneity of tau pathology and how it associates with multifactorial changes in AD, explaining

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discordant awareness of problems and potentially leading to a divergent response to therapy. In ADNI, we found that the clinical high-tau subtype (labeled as S2) exhibited stronger temporal-limbic and frontal involvement and worse cognition than the low-tau subtypes (labeled as S1 and S3), with S3 showing greater proportional tau involvement in medial temporal regions whereas S1 showed greater proportional tau involvement in sensorimotor and frontal regions. Spatially distinct tau patterns in S2 also accompanied discordant awareness of problems, not seen in S1 or S3.

Emily Hector (University of Michigan)

A new block covariance regression model and inferential framework for massively large neuroimaging data

Some evidence suggests that people with autism spectrum disorder exhibit patterns of brain functional dysconnectivity relative to their typically developing peers, but specific findings are difficult to replicate. To facilitate this replication goal with data from the Autism Brain Imaging Data Exchange (ABIDE), we propose a flexible and interpretable model for participant-specific voxel-level brain functional connectivity. Our approach efficiently handles massive participant-specific whole brain voxel-level connectivity data that exceed one trillion data points. The key component of the model is to leverage the block structure induced by defined regions of interest to introduce parsimony in the high-dimensional connectivity matrix through a block covariance structure. Associations between brain functional connectivity and participant characteristics—including eye status during the resting scan, sex, age, and their interactions—are estimated within a Bayesian framework. A spike-and-slab prior facilitates hypothesis testing to identify voxels associated with autism diagnosis. Simulation studies are conducted to evaluate the empirical performance of the proposed model and estimation framework. In ABIDE, the method replicates key findings from the literature and suggests new associations for investigation.

Paul Taylor (National Institute of Mental Health)

Go Figure: Transparency in neuroscience images preserves context and clarifies interpretation

Historically, neuroimaging results and figures only include regions that surpass a given statistical threshold. However, this common step strongly biases interpretations and meta-analyses, particularly towards non-reproducibility. We advocate instead for "transparent thresholding," which highlights statistically significant regions but also includes subthreshold locations for key experimental context. This balances distilling modeled results with enabling informed interpretations. We highlight transparent thresholding's many benefits with several examples and real-world datasets. We also bring attention to the many software packages that have now implemented it. Compared to standard results reporting, transparent thresholding removes ambiguity, decreases hypersensitivity to non-physiological features, catches potential artifacts, improves cross-study comparisons, reduces non-reproducibility biases, and clarifies interpretations.

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Track B · S2

Advances in methods for cancer imaging: From single-cell data to sustainable AI

Organizer: Andrew Whiteman · Chair: Tim Johnson

Inna Chervoneva (Thomas Jefferson University)

Spatial index functional protein biomarkers based on single-cell imaging data

Abstract to be announced

Lubomir Hadjiski (University of Michigan)

Integrated approach: Large language models, deep learning and radiomics for survival prediction of bladder cancer patients
Advances in natural language processing (NLP) and machine learning could assist human users in clinical data extraction from unstructured electronic medical records (EMRs). The potential applications of large language models (LLMs) in healthcare are vast, from automating routine tasks to supporting complex decision-making processes. For instance, in oncology, where patient records often contain extensive unstructured information, LLMs could streamline workflows by extracting and organizing relevant data for further use with imaging and/or clinical evaluation. Furthermore, the extracted data could be integrated rapidly into predictive models for more robust assessments of patient prognosis. We have evaluated the performance consistency of an LLM-aided survival prediction model that combined clinical, radiomic, and deep learning (CRD) information, for 5-year survival prediction of bladder cancer patients. The predictive CRD model was based on clinical descriptors extracted by LLM from the medical records, and radiomics and deep-learning features extracted from computer tomography images. The clinical and imaging information were fused to provide survival prediction. The combined CRD model has improved the prediction of the 5-year survival of bladder cancer patients. The integrated LLM-CRD model demonstrated robust performance in predicting patient survival. Accurate survival prediction for bladder cancer patients can provide valuable information for treatment planning, patient counseling, decision-making, and resource allocation.

Andrew Whiteman (University of Michigan)

Using computer vision-inspired kernels to predict cancer prognosis from multiplexed images of tumor micro-environments
Spatial multiplex imaging has been given considerable interest by the cancer research community in recent years. This technique can simultaneously quantify the expression of multiple protein markers in the same sample of tissue. The technology promises to improve characterization of tumor microenvironments and ultimately facilitate quantification of disease progression or prognosis, response to therapy, etc. Images of tumor biopsy slides are typically processed and resolved at the single cell-level, where each cell is labeled with protein maker intensity information. The analytical challenge is then how best to make use of the resulting wealth of spatially-dependent biomarker information. Here, we propose a scalar-on-image modeling framework using computer-vision

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inspired kernels. This framework identifies morphological anchors within each image using graph convolutions on local cell networks, and allows us to quantify similarity between biopsies based on local patterns of marker expression. We showcase the utility of this approach inferring factors that relate to cancer prognosis in real-world data.

Shu (Joy) Jiang (Washington University in St. Louis)

Sustainable AI in cancer imaging: From research to market

The translation gap between published research and clinical impact is severe: tens of thousands of AI and statistical papers on medical imaging are published annually, yet fewer than 0.1% reach patients. This talk examines why methodological rigor alone does not drive market adoption. Through case studies of research projects, from uncertainty quantification to domain adaptation, we identify the disconnect between academic validation and clinical requirements. Key insights include the importance of early clinician engagement, the divergence between statistical metrics and clinical outcomes, and the regulatory and validation frameworks necessary for commercialization. We present the complete pathway from research to market exit, including regulatory strategy, prospective validation design, and go-to-market mechanics. The evidence demonstrates that sustainable AI requires integration of scientific rigor, clinical understanding, and business viability from the outset.

9:30–10:50 AM

Track A · S3

AI for Medical Image Analysis I

Organizer: Kayvan Najarian · Chair: Kayvan Najarian

Xiaoxing Ren (Cornell University)

A Communication-Efficient Decentralized Learning Algorithm for Multi-Center Medical Image Classification

Multi-center medical image analysis is limited by privacy, heterogeneous data, and high communication costs. We propose a communication-efficient decentralized learning framework where each medical site performs local updates and only exchanges model parameters. The method maintains privacy, greatly reduces communication, and achieves accuracy comparable to centralized training on multi-institution classification tasks.

John Reavey-Cantwell & Emily Wittrup (Virginia Commonwealth University & University of Michigan, Ann Arbor)

An Integrative AI Method for Stroke Treatment Assessment and Management

With variety of imaging and EHR data available in clinical settings, integrative approaches towards management and assessment of surgical procedures for complex complications such as stroke have become a necessity in clinical settings. This talk will discuss a hierarchical AI approach to create such as an integrative solution for stroke management.

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Amoon Jamzad & Parvin Mousavi (Queen's University)

Multimodal Intraoperative Margin Assessment in Breast Conserving Surgery

Improving outcomes in cancer surgery increasingly depends on intelligent tools that can guide real-time decision-making. While many computer-assisted interventions rely on conventional imaging technologies, mass spectrometry—a technique that captures detailed molecular signatures of tissues—is emerging as a powerful alternative modality. When integrated with AI-driven tissue classification models, mass spectrometry can help surgeons accurately distinguish between healthy and cancerous tissues. However, to enable practical intraoperative use, predictions must be localized and communicated effectively to surgeons to support decision-making. In this study, we evaluate the performance of navigated rapid evaporative ionization mass spectrometry (REIMS) as a tool for intraoperative margin assessment in breast cancer surgery.

Robert Policelli & Parvin Mousavi (Queen's University)

A pathway-based approach to improve breast cancer classification explainability

Accurate intraoperative distinction between breast tumour and healthy tissue could improve surgical precision and reduce re-operations for patients requiring breast conserving surgery. While mass spectrometry data can be used to classify breast cancer tissue, its lack of explainability limits clinical implementation. Linking predictions to biological pathways allows clinicians to rapidly understand the classification logic. By mapping mass spectrometry data to biological pathways using publicly available databases, we created concept-based models that attribute specific pathways to the prediction of breast cancer or healthy tissue. With further refinement and validation, this could act as a decision-support tool, combining clinician expertise with rapid, explainable tissue analysis to optimize patient treatment.

Track B · S4

Statistical Innovation in Multiplex Imaging: Methods for Robust Discovery

Organizer: Yue Wang · Chair: Yue Wang

Thao Vu (Colorado School of Public Health)

Spatial and functional analytic approaches for single-cell imaging data using entropy measures

Spatial heterogeneity in the tumor microenvironment (TME) plays a critical role in gaining insights into tumor development and progression. None of the conventional approaches has fully accounted for the simultaneous heterogeneity caused by both cellular diversity and spatial configurations of multiple cell categories. In this paper, we propose an approach to leverage spatial entropy measures at multiple distance ranges to account for the spatial heterogeneity across different cellular organizations. Functional principal component analysis (FPCA) is applied to estimate FPC scores which are used as predictors in a Cox regression model to

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investigate the impact of spatial heterogeneity in the TME on survival outcome, potentially adjusting for other confounders. Using non-small cell lung cancer and triple-negative breast cancer datasets as case studies, we found that the spatial heterogeneity in the TME immune cellular composition had significant effects on the overall survival. Furthermore, simulation studies demonstrated a high predictive power of the proposed method by accounting for both clinical effect and the impact of spatial heterogeneity.

Souvik Seal (The Medical University of South Carolina)

SpaceBF: Spatial coexpression analysis using Bayesian Fused approaches in spatial omics datasets

Advances in spatial omics enable measurement of genes (spatial transcriptomics) and peptides, lipids, or N-glycans (mass spectrometry imaging) across thousands of locations within a tissue. While detecting spatially variable molecules is a well-studied problem, robust methods for identifying spatially varying co-expression between molecule pairs remain limited. We introduce SpaceBF, a Bayesian fused modeling framework that estimates co-expression at both local (location-specific) and global (tissue-wide) levels. SpaceBF enforces spatial smoothness via a fused horseshoe prior on the edges of a predefined spatial adjacency graph, allowing large, edge-specific differences to escape shrinkage while preserving overall structure. In extensive simulations, SpaceBF achieves higher specificity and power than commonly used methods that leverage geospatial metrics, including bivariate Moran's I and Lee's L. We also benchmark the proposed prior against standard alternatives, such as intrinsic conditional autoregressive and Matérn priors. Applied to spatial transcriptomics and proteomics datasets, SpaceBF reveals cancer-relevant molecular interactions and patterns of cell-cell communication (e.g., ligand-receptor signaling), demonstrating its utility for principled, uncertainty-aware co-expression analysis of spatial omics data.

Siyuan Ma (Vanderbilt University)

Batch-Agnostic Self-Supervised Learning Uncovers Clinically Relevant Tumor Niches in Multiplexed Imaging

Multiplexed imaging enables rich, high-resolution characterization of the tumor microenvironment but relies on labor-intensive and error-prone cell segmentation and phenotyping pipelines. We present BASSL-MI, a batch-agnostic, self-supervised framework for discovering tissue niches directly from multiplexed imaging data. BASSL-MI operates directly on image patches, eliminating the need for explicit cell segmentation while mitigating image-, sample-, or batch-specific artifacts. Built on a modified contrastive block disentanglement architecture, BASSL-MI learns dual latent representations that separate biologically informative features from batch-dependent factors through spatially guided augmentations and batch-invariance objectives. Applied to a 56-marker colorectal cancer CODEX dataset, BASSL-MI-trained embeddings markedly reduce image-specific variability and recover biologically interpretable spatial niches. Notably, it uncovers CD20-rich follicular regions associated with improved survival, outperforming published findings from cell segmentation-driven clustering.

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Ruoqian Liu (Weill Cornell Medical College)

Detecting Disease-Associated Dysbiosis in Image-Based Spatial Microbiome Data

Spatial metagenomic sequencing has opened a new frontier for characterizing microbial communities directly within their native tissue environments, generating high-resolution biofilm maps that richly complement existing spatial multi-omics data. Despite the growing availability of such data, analytical methods that fully exploit the spatial organization of microbial profiles to identify disease-associated microbial dysbiosis remain scarce. We present a novel statistical framework for detecting differential spatial patterns in microbial communities between healthy and diseased groups using image-based spatial microbiome data. By explicitly modeling the spatial arrangement of microbial profiles, our method captures biologically meaningful alterations that cannot be detected by composition-only analyses. Extensive simulations demonstrate high sensitivity, robustness, and adaptability of our method across diverse and noisy spatial architectures. When applied to an oral biofilm dataset, the method uncovers microbial patches that strongly correlate with disease status, providing actionable biological insights.

Track C · S5

Alignment, Calibration, and Robust Inference in Neuroimaging

Organizer: Panpan Zhang · Chair: Panpan Zhang

Zhengwu Zhang (University of North Carolina at Chapel Hill)

Alignment of Continuous Brain Connectivity

Brain networks are typically represented by adjacency matrices, where each node corresponds to a brain region. In traditional brain network analysis, nodes are assumed to be matched across individuals, but the methods used for node matching often overlook the underlying connectivity information. This oversight can result in inaccurate node alignment, leading to inflated edge variability and reduced statistical power in downstream connectivity analyses. To overcome this challenge, we propose a novel framework for registering high resolution continuous connectivity (ConCon), defined as a continuous function on a product manifold space—specifically, the cortical surface—capturing structural connectivity between all pairs of cortical points. Leveraging ConCon, we formulate an optimal diffeomorphism problem to align both connectivity profiles and cortical surfaces simultaneously. We introduce an efficient algorithm to solve this problem and validate our approach using data from the Human Connectome Project (HCP). Results demonstrate that our method substantially improves the accuracy and robustness of connectome-based analyses compared to existing techniques.

Yiwen Dong (University of Pittsburgh)

Alzheimer's Disease Biomarkers Harmonization Challenges

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Recent studies have shown that Alzheimer's disease imaging biomarkers remained highly variable even after state-of-the-art harmonization methods application. Furthermore, data harmonization and standardization across different clinical domains including Positron Emission Tomography (PET) imaging and blood biomarkers continue to hinder our ability to accurately estimate differences across different clinical groups. In this study we present different harmonization methods for PET imaging and blood biomarkers outcomes in Alzheimer's Disease and aim to explain the variability across several harmonization methods.

Andrew Chen (Medical University of South Carolina)

Calibration of MRI-Based Reference Intervals to New Samples

Reference intervals, defined as intervals containing a new observation with a specified probability relative to reference data, would be clinically useful in assessing brain magnetic resonance imaging (MRI). Brain charts, which are estimates of MRI phenotypes across covariates such as age and sex, can be used to construct reference intervals. However, the reference data used to fit intervals often differs from a new sample in terms of study design, MRI acquisition, and image preprocessing. Application of MRI reference intervals to new samples remains a challenging problem. Here, we propose a new method called Reference interval calibration via conformal prediction (ReForm) that adjusts reference intervals for a new sample. Our method builds on recent work in conformal prediction, which yields intervals with guaranteed coverage for new observations. Through resampling experiments in Lifespan Brain Chart Consortium cortical thickness data, we compare ReFormed reference intervals to refitting intervals, statistical harmonization methods, and model-based adjustment of intervals. Notably for patient privacy concerns, ReForm does not require sharing of reference data. Yet, our empirical results demonstrate that ReForm controls FPR similarly or better than alternative methods which require sharing reference data. Finally, we provide recommendations for practical applications of ReForm and an R package for calibrating reference intervals using ReForm.

Xinyu Zhang (Vanderbilt University)

Asymptotic Distribution of Robust Effect Size Index

The Robust Effect Size Index (RESI) is a recently proposed standardized index to quantify effect magnitude across models, with significant applications in high-dimensional data analysis, including neuroimaging studies. However, its confidence interval (CI) construction has relied on computationally intensive bootstrap procedures. We establish a general theorem for the asymptotic distribution of the RESI using a Taylor expansion that accommodates a broad class of models. Simulations under various linear and logistic settings show that RESI and its CI have smaller bias and more reliable coverage than commonly used effect sizes such as Cohen's d and f , and that combining with robust covariance estimation yields valid inference under model misspecification—a crucial aspect in neuroimaging, where model assumptions often need to account for complex spatial dependencies and heterogeneous variances across regions. Additionally, our approach shows that the asymptotic method significantly reduces

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computational time, achieving up to a 50-fold speedup over the bootstrap procedure. This is particularly valuable in neuroimaging applications where large-scale datasets and computational efficiency are key challenges.

11:00–11:15 AM

Break & Welcome Address

11:15 AM–12:15 PM

Keynote 1 (11:30 AM): Prof. Hernando Ombao (KAUST) (Michigan League Ballroom)

Spectral Measures of Brain Functional Connectivity

This talk will present a broad range of statistical methods for characterizing dependence in a brain network. First, we present a general framework which decomposes each signal into various frequency components and then characterize the dependence properties through these oscillatory activities. Some of the most prominent frequency domain measures such as coherence, partial coherence, and dual-frequency coherence can be derived as special cases under this general framework.

In the second part of the talk, we will briefly describe current work on information-theoretic measures of connectivity. This class of measures are based on joint (and conditional) densities rather than merely the first two moments. The proposed spectral transfer entropy (STE) quantifies the magnitude and direction of information flow from a certain frequency-band oscillation of a channel to an oscillation of another channel. A novel contribution is a simple, yet efficient estimation method based on vine copula theory that enables estimates to capture zero (boundary point) without the need for bias adjustments. With the vine copula representation, a null copula model, which exhibits zero STE, is defined, making significance testing for STE straightforward through a standard resampling approach.

In the third part of the talk, we propose a new spectral dependence measure, the nonlinear vector coherence (NVC), to capture beyond-linear interactions between oscillations from two distinct brain regions (each contains many channels). This addresses the limitations of conventional pairwise channel-to-channel approaches and defines a more natural region-to-region connectivity framework in the frequency domain. We further introduce a rank-based inference procedure that enables fast and distribution-free estimation of the proposed measure, as well as a fully nonparametric test for spectral independence. The empirical performance of our proposed inference methodology is demonstrated through extensive numerical experiments. An application to resting-state EEG data reveals that our novel NVC measure uncovers distinct and diagnostically meaningful connectivity patterns which effectively discriminate healthy individuals from those with Alzheimer's Disease and Fronto-Temporal Dementia.

This is joint work with Paolo Redondo and Raphael Huser at KAUST.

12:15–1:30 PM

Lunch (12:30 PM)

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1:30–2:50 PM

Track A · S6

Spatiotemporal Models for Brain Structure, Function, and Connectivity

Organizer: Mark Fiecas · Chair: Tim Johnson

Mara Sherlin Talento (KAUST)

KenCoh: A Rank-Based Estimator of Coherence between Multivariate Time Series

This work is inspired by the problem of characterizing a dependence measure between two cortical regions of the brain where each region contains multiple signal recordings from several neurons or channels (e.g., inhibitory and excitatory neurons). The goal is to identify differences in the structure of brain functional connectivity between known brain states. An exploratory tool for studying the dependence between two random vectors is via canonical correlation analysis. However, these are limited to only capturing linear associations and are sensitive to outlier observations. Mitigating these limitations is crucial because brain functional connectivity is likely to be more complex than linear, and brain signals may exhibit heavy-tailed properties. To overcome these limitations, we develop a robust method, Kendall's tau-based canonical coherence (KenCoh), to learn connectivity structure among neuronal signals filtered at given frequency bands. Our simulation study demonstrates that KenCoh is competitive with the moment-based estimator and outperforms the latter when the underlying distributions are heavy-tailed. We apply our method to EEG recordings from a virtual-reality driving experiment and to calcium imaging recordings in inhibitory and excitatory neurons of the auditory cortex in mice subjected to sound stimuli. Our findings reveal distinct regional dependencies across frequency bands and brain states.

Yuhyeong Jang (Southern Methodist University)

A Survival Analysis of Glioma Patients using Topological Features

Tumor shape contains critical information regarding both the growth and metastasis of gliomas. To explore the potential prognostic value of tumor shape for survival outcomes, we introduce a novel imaging feature derived from brain MRI scans that topologically characterizes tumor morphology using persistent homology. This feature captures shape patterns that are not accounted for by conventional radiomic measures. To incorporate the proposed feature into statistical modeling of patients' survival outcomes, we represent it as functional data and integrate it into a functional Cox regression model. We further investigate lobe-specific effects by including interaction terms between the proposed feature and tumor location.

In a case study of 133 glioblastoma patients, the prognostic utility of the proposed approach in a clinical setting is evaluated through its predictive performance for risk stratification. The study findings suggest that the topological features are strong predictors of survival outcomes, remaining significant after adjustment for clinical variables and providing additional clinically meaningful insights into tumor behavior.

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Heather Shappell (Wake Forest University)

A Hidden Semi-Markov Model Framework for Covariate-Dependent Sojourn Distributions in Dynamic Brain Network Analysis

The study of functional brain networks increasingly emphasizes how connectivity patterns evolve over time. While hidden Markov models (HMMs) have become a popular tool for modeling dynamic functional connectivity (FC), their assumption of geometrically distributed sojourn times (consecutive time spent in a state) forces unrealistically frequent state switching. Hidden semi-Markov models (HSMMs) address this limitation by explicitly modeling the sojourn distribution, enabling more accurate characterization of the duration spent in each network state. I develop an HSMM framework for inferring dynamic brain network states from functional MRI data and estimating subject-specific state sequences, dwell times, and associated connectivity patterns. Building on this foundation, I introduce an extension in which the sojourn distribution is modeled as a function of covariates. This covariate-dependent formulation allows state dwell times to vary systematically with demographic, clinical, or behavioral factors, providing a direct mechanism for comparing temporal dynamics across patient populations. I illustrate the utility of this approach using fMRI data from older adults with obesity, demonstrating how covariate effects on sojourn behavior can reveal meaningful differences in network stability and flexibility. This framework opens new avenues for studying how individual characteristics shape the temporal organization of functional brain networks.

Joshua Lukemire (Emory University)

Hierarchical Modeling of Localized Intracranial Volume Abnormalities in Craniosynostosis

Craniosynostosis, the premature fusion of cranial sutures, alters skull morphology and can restrict localized brain development despite normal overall intracranial volume. These localized deformations are linked to long-term neurodevelopmental deficits, but there is a current lack of quantitative models that capture the heterogeneity in growth patterns driven by different suture fusion types. Leveraging a large dataset of CT and 3D photogrammetry scans from craniosynostosis patients and healthy controls, a novel hierarchical generalized additive model that uses soap-film smooths is proposed, and basis functions are shared to model continuous volume maps across space, age, and fusion type. This approach enables fine-scale localization of volume abnormalities and prediction of future development trajectories. The model accounts for covariates such as age and sex and is estimated using a Bayesian framework.

Track B · S7

Advanced Statistical and Machine Learning based methods for Spatial imaging and omics

Organizer: Satwik Acharyya · Chair: Andrew Whiteman

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Satwik Acharyya (University of Alabama at Birmingham)

Network models for Spatial omics data

Network models are powerful tools to investigate complex dependence structures in high throughput genomic datasets. They allow for holistic, systems-level view of the various biological processes, for intuitive understanding and coherent interpretations. However, most existing network or graphical models are developed under assumptions of homogeneity of samples and are not readily amenable to modeling spatial heterogeneity which often manifests in spatial genomics data. In this talk, I will discuss two spatial network models focusing on spatially varying covariance and precision matrices. (I) SpaceX (spatially dependent gene co-expression network) is a Bayesian methodology to identify both shared and cluster-specific co-expression networks across genes. (II) Spatial Graphical Regression (SGR) is a flexible approach based on graphical regression that enables spatially varying graphs over the spatial domain of the tissue. The framework incorporates multiple spatial covariates and provides a linear and non-linear functional mapping between the spatial domain and the precision matrices. All the approaches are illustrated by using case studies from cancer genomics.

Ronglai Shen (Memorial Sloan Kettering Cancer Center)

Spatial Immunophenotyping from Whole-Slide Multiplexed Tissue Imaging using Convolutional Neural Networks

The multiplexed immunofluorescence (mIF) platform enables biomarker discovery through simultaneous detection of multiple markers on a single tissue slide, offering detailed insights into intratumor heterogeneity and the tumor-immune microenvironment at spatially resolved single cell resolution. However, current mIF image analyses are labor-intensive, requiring specialized pathology expertise which limits their scalability and clinical application. To address this challenge, we developed CellGate, a deep-learning (DL) computational pipeline that provides streamlined, end-to-end whole-slide mIF image analysis including nuclei detection, cell segmentation, cell classification, and combined immuno-phenotyping across stacked images. The model was trained on over 2 million single cell images from 34 melanomas in a retrospective cohort of patients using whole tissue sections stained for an immune marker panel with manual gating and extensive pathology review. When tested on new whole mIF slides, the model demonstrated high precision-recall AUC. Further validation on whole-slide mIF images of primary melanomas from an independent cohort confirmed that CellGate can reproduce expert pathology analysis with high accuracy. We show that spatial immuno-phenotyping results using CellGate provide deep insights into the immune cell topography and differences in T cell functional states and interactions with tumor cells in patients with distinct histopathology and clinical characteristics. This pipeline offers a fully automated and parallelizable computing process with substantially improved consistency for cell type classification across images, potentially enabling high throughput whole-slide mIF tissue image analysis for large-scale clinical and research applications.

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Michele Peruzzi (University of Michigan)

Integrated Geospatial Network models for Spatial Omics

Multiplexed imaging for spatial cancer omics measure dozens to hundreds of molecular and cellular features across tissues, producing spatially structured data. Network models are attractive because they aim to distinguish direct from mediated interactions among cell types and molecular markers. However, these measurements exhibit strong local spatial correlation; without a spatial process model, network estimates can confound direct interaction with simple spatial smoothness and co-localization.

In this talk, we introduce methods that bridge geospatial modeling of spatial dependence via multivariate Gaussian processes with Gaussian graphical models for conditional independence. Our Integrated Geospatial Network (IGN) models simultaneously capture spatial correlation and enable inference on process-level conditional independence. This is conditional independence across infinite-dimensional spatial surfaces rather than point-wise conditional independence at individual locations. The resulting network-of-processes interpretation can separate direct from mediated dependence while correctly accounting for spatial structure. IGN's joint interpretation is especially important for identifying candidate interaction pathways in cancer: IGNs can distinguish direct spatial links between markers or cell types from associations driven by shared tissue context or mediating processes such as vasculature or stroma.

Veera Baladandayuthapani (University of Michigan)

Bayesian Statistical Models for Spatial Imaging in Cancer

Spatial biology plays a crucial role in understanding the tumor microenvironment (TME) and is of immense importance in cancer research. Broadly, the TME is a complex biological ecosystem consisting of various cell types, signaling molecules, extra-cellular components, and spatial organization. Modern techniques such as spatial transcriptomics and multiplex immunofluorescence imaging facilitate the assessment, characterization, and visualization of various aspects of molecular spatial heterogeneity within the TME. In this talk, I will present statistical methods to answer some relevant scientific questions of interest for such data including spatial varying genomic networks and transcriptional programs, quantification of inter-cellular interactions in the TME and their association with patient-specific outcomes. The focus will be on probabilistic formulations based on effective dimension reduction techniques which allow for rigorous inference and uncertainty quantifications. The methods will be exemplified using several case studies in cancer.

Statistical Methods in Imaging (SMI) 2026 – Program

Track C · S8

Principled Statistical and Machine Learning Methods for Brain Imaging Inference

Organizer: Zhengwu Zhang · Chair: Zhengwu Zhang

Benjamin Risk (Emory University)

Causal inference for hierarchical machine learning with an application to stimulant impacts on brain connectivity in autism
In biomedical research, repeated measurements within each subject are often processed to remove artifacts and unwanted sources of variation. The resulting data are used to construct derived outcomes that act as proxies for outcomes that are not directly observable. Although intra-subject processing is widely used, its impact on inter-subject statistical inference has not been systematically studied, and a principled framework for causal analysis in this setting is lacking. In this article, we propose a semiparametric framework for causal inference with derived outcomes obtained after intra-subject processing. This framework applies to settings with a modular structure, where intra-subject analyses are conducted independently across subjects and are followed by inter-subject analyses based on parameters from the intra-subject stage. We develop multiply robust estimators of causal parameters under rate conditions on both intra-subject and inter-subject models, which allows the use of flexible machine learning. We specialize the framework to a mediation setting and focus on the natural direct effect. For high dimensional inference, we employ a step-down procedure that controls the exceedance rate of the false discovery proportion. Simulation studies demonstrate the superior performance of the proposed approach. We apply our method to estimate the impact of stimulant medication on brain connectivity in children with autism spectrum disorder.

Armin Iraj (Georgia State University)

From Signals to Sources: Advances in Dynamic Precision Functional Mapping

Functional magnetic resonance imaging (fMRI) has transformed our understanding of brain organization; however, conventional analysis pipelines often rely on anatomically fixed regions of interest that fail to adequately capture the dynamic and individualized nature of brain function. Although data-driven approaches estimate subject-specific intrinsic connectivity networks and improve individual-level representation, they typically continue to assume static spatial patterns. In this lecture, I present recent advances in Dynamic Precision Functional Mapping, a framework that moves beyond conventional approaches by capturing individualized, time-resolved functional sources through mathematically grounded, data-driven modeling. Furthermore, I will highlight recent efforts to incorporate nonlinear representations and interactions to enhance the fidelity of individualized mapping. Finally, I will demonstrate the utility of these methods in studying psychosis.

Guorong Wu (UNC Chapel Hill)

Uncover Governing Law of Pathology Propagation Mechanism Through A Mean-Field Game

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Alzheimer's disease (AD) is marked by cognitive decline along with the widespread of tau aggregates across the brain cortex. Due to the challenges of imaging pathology spreading flows in-vivo, however, quantitative analysis on the cortical pathways of tau propagation and its interaction with the cascade of amyloid-beta ($A\beta$) plaques lags behind the experimental insights of underlying pathophysiological mechanisms. To address this challenge, we present a physics-informed neural network, empowered by mean-field theory, to uncover the biologically meaningful spreading pathways of tau aggregates between two longitudinal snapshots. Following the notion of 'prion-like' mechanism in AD, we first formulate the dynamics of tau propagation as a mean-field game (MFG), where the spread of tau aggregate at each location (aka. agent) depends on the collective behavior of the surrounding agents as well as the potential field formed by amyloid burden. Given the governing equation of propagation dynamics, MFG reaches an equilibrium that allows us to model the evolution of tau aggregates as an optimal transport with the lowest cost in Wasserstein space. By leveraging the variational primal-dual structure in MFG, we propose a Wasserstein-1 Lagrangian generative adversarial network (GAN), in which a Lipschitz critic seeks the appropriate transport cost at the population level and a generator parameterizes the flow fields of optimal transport across individuals. Additionally, a symbolic regression module derives an explicit $A\beta$ -tau crosstalk formula. Experimental results on public neuroimaging datasets demonstrate that our explainable deep model yields accurate tau predictions for unseen subjects and offers new insights into AD pathology spread.

Will Consagra (University of South Carolina)

Learned Hemodynamic Coupling Inference in Resting-State Functional MRI

Functional magnetic resonance imaging (fMRI) provides an indirect measurement of neuronal activity via hemodynamic responses that vary across brain regions and individuals. Ignoring this hemodynamic variability can bias downstream connectivity estimates. Furthermore, the hemodynamic parameters themselves may serve as important imaging biomarkers. Estimating spatially varying hemodynamics from resting-state fMRI (rsfMRI) is therefore an important but challenging blind inverse problem, since both the latent neural activity and the hemodynamic coupling are unknown. In this work, we propose a methodology for inferring hemodynamic coupling on the cortical surface from rsfMRI. Our approach avoids the highly unstable joint recovery of neural activity and hemodynamics by marginalizing out the latent neural signal and basing inference on the resulting marginal likelihood. To enable scalable, high-resolution estimation, we employ a deep neural network combined with conditional normalizing flows to accurately approximate this intractable marginal likelihood, while enforcing spatial coherence through priors defined on the cortical surface that admit sparse representations. Uncertainty in the hemodynamic estimates is quantified via a double-bootstrap procedure. The proposed approach is extensively validated using synthetic data and real fMRI datasets, demonstrating clear improvements over current methods for hemodynamic estimation and downstream connectivity analysis.

3:00–4:20 PM

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Statistical Methods in Imaging (SMI) 2026 – Program

Track A · S9

Advanced statistical and AI models for complex and graphical brain images

Organizer: Yize Zhao · Chair: Yize Zhao

Yuexuan Wu (University of South Carolina)

High-Dimensional Multivariate Mediation Analysis for Brain Imaging via Structured Dimension Reduction

Causal mediation analysis is critical for understanding how changes in the brain mediate the effects of environmental and genetic factors on neurological outcomes in neuroimaging studies. However, traditional mediation methods often face challenges when mediators are high-dimensional structured objects, such as complex brain imaging data, due to the curse of dimensionality and reduced statistical power. This study introduces a methodology that leverages envelope-based structured dimension reduction to enhance pathway identification and statistical power in detecting indirect effects. The proposed approach is applied to Alzheimer's Disease Neuroimaging Initiative (ADNI) data to examine how structural changes in brain regions mediate the impact of genetic factors on cognitive decline. Simulation studies validate the asymptotic properties of the estimators and demonstrate that the method outperforms existing techniques in improved power and reduced variance in estimation. This approach advances mediation analysis for neuroimaging and extends to other high-dimensional multivariate contexts, offering a robust framework for disease detection and intervention strategies.

Zhihao Wu (Yale University)

Dependence-Aware Knockoff Enables Stable Feature Discovery in Brain Imaging

In neuroscience, high dimensionality and complex dependence structures often lead to substantial variability across variable selection methods, even when predictive performance is comparable. We propose a knockoff framework with a novel generation strategy that extends the conventional sequential conditional independent pairs (SCIP) algorithm through conditional diffusion modeling and feature partitioning, enabling computationally efficient construction while preserving complex dependence patterns and spatial contiguity in high-dimensional data. Our approach is general and flexible, accommodating diverse data types and supporting both unimodal and multimodal settings, while rigorously controlling the false discovery rate (FDR) and producing robust inference. We apply the method to investigate behavioral outcomes using vectorized structural MRI features, matrix-valued functional connectivity from fMRI, and their joint representations. Across feature importance measures derived from multiple scalar-on-image models, our framework yields stable and consistent variable selection results, demonstrating enhanced robustness and interpretability for imaging-based inference.

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Qiong Wu (University of Pittsburgh)

Distributed Covariance Graph-Guided ComBat (dG-ComBat): A Privacy-Preserving Framework for Harmonizing Multi-Site Neuroimaging Data

Multi-site neuroimaging studies enhance statistical power and generalizability but face challenges from data privacy constraints where direct data sharing is prohibited. In addition, neuroimaging data across multiple institutions often exhibit systematic heterogeneity driven by scanner differences, which could bias downstream analyses if not addressed. We propose distributed covariance graph-guided ComBat (dG-ComBat), a multivariate harmonization framework that operates in a fully privacy-preserving distributed setting. dG-ComBat identifies and leverages the latent covariance graph structure among neuroimaging features to guide mitigation of both mean and covariance batch effects. With only two rounds of summary-statistics exchange, dG-ComBat achieves identical harmonization performance (i.e., lossless) to centralized setting in which data from all participating institutions are shared. Extensive simulations and applications to amyloid- β and tau PET data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) showed that dG-ComBat effectively reduces scanner-related variability while preserving biologically relevant diagnostic signals, and improving downstream analyses. The method provides a scalable, summary-statistics-based approach for harmonizing multi-institutional neuroimaging data when individual-level data sharing is infeasible.

Xinyuan Tian (Yale University)

Estimating a shared backbone linking structural and functional connectomes via bidirectional graph autoencoders
Structural connectivity (SC) derived from diffusion MRI and functional connectivity (FC) derived from functional MRI provide complementary summaries of brain organization. Because coherent brain function is supported by anatomically grounded, polysynaptic communication pathways, we model SC and FC as arising from a shared, biologically meaningful scaffold and propose a bidirectional graph autoencoder to estimate a latent backbone common to both modalities. The method fits coupled forward and reverse mappings (SC $\rightarrow$ FC and FC $\rightarrow$ SC) that share a single latent adjacency matrix, which serves as the propagation operator for message passing in both modalities and yields an explicit backbone estimate. To account for heterogeneity across subjects and modalities, we augment the shared backbone with modality specific deviations that adapt subject-level topology while remaining anchored to the common foundation. We characterize the estimator via a walk-algebra representation and show that the regularized bidirectional objective recovers a minimal shared backbone under identifiability conditions. In simulations, the proposed approach improves bidirectional reconstruction and more accurately recovers the ground-truth backbone than competing unidirectional and higher-order graph neural baselines. An application to Human Connectome Project (HCP) and Adolescent Brain Cognitive Development (ABCD) data yields reproducible backbone networks that align with established large-

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scale functional systems. Moreover, backbone estimates show moderate cross-cohort consistency between HCP and ABCD, illustrating the utility of the proposed framework for interpretable structure–function modeling.

Track B · S10

Emerging Statistical Techniques for Longitudinal and Multimodal Imaging Analysis

Organizer: Sharmistha Guha · Chair: Ivo Dinov

Ivo Dinov (Statistics Online Computational Resource, Computational Medicine & Bioinformatics, University of Michigan)

[Kime-Phase Tomography and Manifold Representation of Longitudinal Processes over a Complex-Time Domain](#)

Variability in repeated measurement longitudinal processes, such as fMRI, is subject to intrinsic (domain space) latent phase states stochasticity and extrinsic (range space) chronological time dispersion typically modeled as additive noise. I'll discuss complex-time (kime) representation of time-varying processes as parametric manifolds and kime-phase tomography (KPT), a mathematical statistics framework that extends the representation of longitudinal observations from classical time to kime. The kime phase tracks the intrinsic kime-phase domain stochasticity, due to cross-sectional variation in the repeated sampling. The phase is complementary to the classical extrinsic additive noise in the signal range space. The probabilistic formulation of the kime manifold induces different metrics and a family of non-commutative operators that satisfy canonical commutation relations analogous to quantum mechanics. This operator-theoretic foundation enables tomographic reconstruction of latent phase distributions from mixed-moment statistics via spectral deconvolution on the circle. We will show some examples related to time-varying medical images.

Rajarshi Guhaniyogi (Department of Statistics, Texas A&M University)

DeepSpFactor: A Statistically Trustworthy Deep Learning Framework for Spatial Factor Models in Multi-subject Multi-modal Imaging

The rapid growth of high-resolution imaging across multiple subjects and imaging modalities presents remarkable opportunities to advance our understanding of complex biological and neurological processes. However, jointly analyzing these massive, multi-modal imaging datasets on a large number of subjects poses major challenges due to their extreme dimensionality, intricate spatial relationships, and heterogeneous sources. In this talk, I will introduce DeepSpFactor, a novel, interpretable deep learning framework for large-scale, multi-subject, multi-modal imaging integration. DeepSpFactor leverages spatial factor models, powered by deep neural networks (DNNs), to flexibly capture nonlinear and multi-level spatial patterns in imaging data. Our approach dissects image variation into meaningful latent components, models how these patterns relate to subject-specific traits, and robustly accounts for complex spatial dependencies. By combining Bayesian inference with scalable DNN-based computational architecture, DeepSpFactor provides reliable uncertainty quantification, supports the imputation of missing modalities, and

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delivers scientifically interpretable factors that distinguish shared from modality-specific structures, and offers rapid computation. This work opens new possibilities for statistically rigorous and computationally efficient analysis of large imaging studies, and I will discuss both methodological foundations and applications to real-world neuroscience data.

Simon Vandekar (Department of Biostatistics, Vanderbilt University Medical Center)

Semiparametric confidence sets for a robust effect size in Neuroimaging

The majority of neuroimaging inference focuses on hypothesis testing rather than effect estimation. With concerns about replicability, there is growing interest in reporting standardized effect sizes from neuroimaging group-level analyses. Confidence sets for effect sizes were recently developed for neuroimaging, but are restricted to simple univariate contrasts (e.g., one-sample or two-sample test Cohen's d) and cross-sectional data. Thus, existing methods exclude increasingly common longitudinal associations of biological brain measurements with potentially nonlinear, inter- and intra-individual variations in diagnosis, development, or symptoms. We use modern methods for confidence sets combined with a recently proposed robust effect size index to provide a very general approach and software for effect size confidence set inference in neuroimaging. These confidence sets identify regions of the image where the lower or upper simultaneous confidence interval is above or below a given threshold with high probability. We evaluate the coverage and interval width of the proposed procedures using realistic simulations and perform longitudinal analyses of aging and diagnostic differences of cortical thickness in Alzheimer's disease. This comprehensive approach, along with the visualization functions integrated into the pbj R package, offers a robust tool for analyzing repeated neuroimaging measurements.

Sharmistha Guha (Department of Statistics, Texas A&M University)

Multimodal Neuroimaging in Aging: Joint Learning of Network Data and Spatially Correlated Node Attributes

This talk presents a flexible, predictor-dependent joint modeling framework designed for analyzing multimodal neuroimaging data in aging. The framework accommodates network data from multiple subjects over a shared set of spatial nodes and incorporates spatially correlated attributes. It permits simultaneous inference on nodes associated with predictors, spatial relationships of nodal attributes, and predictor-attribute regression relationships. Simulations demonstrate its superior performance by accounting for both network and spatial data correlations. The method is used to analyze data from the Lifespan Cognitive and Motor Neuroimaging Laboratory, integrating structural and functional MRI data to analyze brain connectivity and region-specific attributes. It includes aging-related features and ROI spatial locations to enhance understanding of the interactions between brain structure, function, aging, and other predictors. As a Bayesian approach, it provides uncertainty quantification, delivering robust results even with small samples.

Statistical Methods in Imaging (SMI) 2026 – Program

Track C · S11

Modern Statistical Methods for Neuroimaging

Organizer: Xi Luo · Chair: Yi Zhao

Dr. John Kornak (University of California, San Francisco)

An MCMC Approach to Bayesian Image Analysis in Fourier Space

Bayesian image analysis provides a powerful approach to enhance image quality and emphasize key features by integrating prior knowledge with probabilistic modeling. This talk will describe a Markov chain Monte Carlo approach for Bayesian Image Analysis in Fourier Space (MCMC-BIFS), to enhance analysis efficiency and flexibility.

By broadening the kinds of prior knowledge that can be, and doing so efficiently, BIFS opens up new possibilities for Bayesian image analysis. I'll describe how BIFS takes advantage of frequency-space structure to support rich prior models, and how these can be used in medical imaging applications, including functional Magnetic Resonance Imaging (fMRI) for understanding brain activation patterns.

Dr. Moo Kyung Chung (University of Wisconsin, Madison)

Causality without DAG

Human brain networks exhibit dense recurrent interactions that are poorly represented by directed acyclic graph (DAG) assumptions or edgewise analyses alone. We introduce a variational framework for causal inference without acyclicity, where directed neural interactions are modeled as flows on a simplicial complex and evolved under an energy-minimizing dynamical system. This separates transient components from persistent harmonic flows, which define a low-dimensional Hilbert space of stable directed cycles. By treating cycles as primary objects of inference, the framework allows recurrent organization to be projected, compared, and statistically analyzed as a global geometric object rather than as isolated edges, paths, or local motifs. Applied to resting-state fMRI, the method reveals densely interacting but stable cyclic structures that reconfigure over time while remaining within the same cyclic subspace. These results suggest that dense cyclic interaction is a fundamental organizing principle of human brain networks and provide a scalable vector-space framework for statistical inference on recurrent dynamics in high-dimensional systems. The talk is based on arXiv:2604.17151.

Dr. Dayu Sun (Indiana University)

Sparse Partial Generalized Tensor Regression With Application to Neuroimaging Data

Tensor data, often characterized as multidimensional arrays, have become increasingly prevalent in biomedical studies, particularly in neuroimaging applications. Analyzing these complex datasets can be challenging due to the high dimensionality and inherent structures within tensors. In this work, we propose the Sparse Partial Generalized Tensor Regression (SPGTR) method for modeling

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diverse types of responses, including continuous, binary, and count data, in terms of both tensor and vector/scalar predictors. Our novel mode-wise penalized manifold optimization techniques enable us to achieve dimension reduction and biologically meaningful sparsity in tensor regression coefficient estimation. We establish the asymptotic properties of the proposed estimation facilitated by the envelope concept. We demonstrate the effectiveness of the SPGTR through extensive simulation studies. An application of SPGTR to a functional magnetic resonance imaging (fMRI) study yields new insights into the association between posttraumatic stress disorder (PTSD) and brain connectivity measures, which are not uncovered by existing methods.

Gang Chen (NIH)

Navigating Correlation, Causation, and Interpretation in Resting-State fMRI

Resting-state fMRI has generated influential insights into large-scale brain organization and contributed to clinically relevant applications, largely through correlation-based measures of cross-regional association in BOLD signals. At the same time, interpreting these statistical associations as reflecting underlying neural interactions requires careful consideration of fundamental methodological constraints.

Here, we distinguish two conceptually distinct but often conflated limitations. The first is measurement distortion: the fMRI signal is an indirect and heterogeneous measurement of neural activity, shaped by neurovascular coupling, physiological processes, and measurement-related factors, which can introduce systematic and incompletely characterized biases into estimated correlations. The second is causal non-identifiability: even if correlations perfectly reflected neural synchrony, the resulting correlation structure would not uniquely determine the underlying neural interactions.

Using causal reasoning, simulations, and analytic arguments, we examine how these constraints affect interpretation. We show that measurement-related biases can distort estimated correlations (e.g., attenuating or inflating associations and affecting group comparisons) and can produce reproducible patterns that do not necessarily reflect underlying neural relationships, highlighting that statistical reliability does not guarantee biological validity. We further demonstrate that graph-theoretic metrics derived from these correlations do not, by themselves, justify mechanistic interpretation.

4:30–6:30 PM

Training Tutorials & Short Courses

Empowering Large Language Models with Statistics: A Practical Tutorial for Medical Imaging

Khashayar Namdar (University of Toronto & Dalhousie University, Canada), Dominik A. Deniffel (Technical University of Munich, Germany) Pascal Tyrrell (University of Toronto)

Large Language Models (LLMs) are increasingly used for analyzing radiology reports and other imaging-associated text. However, their outputs are often treated as opaque or deterministic, despite the fact that LLM behavior is inherently statistical and stochastic.

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This tutorial aims to empower LLM-based workflows with statistics, focusing on variability, uncertainty, and evaluation in medical imaging contexts. The tutorial will adopt a hands-on, example-driven approach using an open-source dataset of 3,024 de-identified brain MRI radiology reports, collected for acute ischemic stroke (AIS) classification. This dataset has been previously evaluated using a rigorous stratified 20-fold nested cross-validation framework, providing a statistically grounded reference point for comparing traditional natural language processing methods, domain-specific embeddings, and LLM-based approaches. Participants will first explore how stochastic sampling parameters, such as temperature, introduce variance in LLM outputs when applied to radiology report classification. We will demonstrate how repeated sampling enables empirical characterization of output distributions, in contrast to deterministic pipelines commonly used in traditional machine-learning workflows. The tutorial will then introduce methods for extracting and interpreting probabilistic signals from LLMs, including log-probabilities and related confidence surrogates, and discuss their role in uncertainty measurement. Using this AIS dataset, we will contrast classical feature-based and embedding-based models, whose performance can be summarized using metrics such as AUC and confidence intervals, with zero-shot LLM inference, where standard performance measures may be unavailable or ill-defined. Visual tools from previous research will be introduced to compare statistical behavior across model classes and across local versus online LLM deployments, highlighting trade-offs in stability and computational load. This tutorial is designed for interdisciplinary researchers focused on applied medical image analytics, including imaging researchers, statisticians, and clinicians. By grounding all concepts in a real radiology report dataset, attendees will gain practical insight into how statistical thinking can improve the evaluation, interpretation, and reliability of LLMs in medical imaging research.

Manifold learning and dimension reduction for imaging data

Chunming Zhang (University of Wisconsin–Madison)

This short-course introduces modern methods for uncovering low-dimensional structure in high-dimensional and complex imaging data, motivated by applications in neuroscience, genetics, biology, and medical research. The course covers key ideas and the mathematical and statistical foundations of dimension reduction, manifold learning, and manifold fitting, together with practical implementation tools. Applications to imaging data, functional data, and single-cell analysis will illustrate how these methods enable more effective dimension reduction, denoising, and representation learning.

Methods for fMRI processing, quality control and group analysis in AFNI

Paul Taylor & Gang Chen (Scientific and Statistical Computing Core, NIMH, NIH)

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fMRI datasets are noisy and typically require many steps of processing before they can be used within a study or clinical practice. We first discuss considerations for data visualization, understanding and processing at the single subject level. This includes setting up a full processing pipeline through regression with `afni_proc.py`. We highlight systematic, useful features in its QC HTML, as well as other scriptable QC tools in AFNI. We then delve into the hierarchical framework of fMRI modeling in AFNI, focusing on four critical aspects of the analytical pipeline: 1) Individual-level modeling: managing the temporal structure of BOLD data by accounting for auto-correlation in the residuals. 2) Group-level generalization: Assessing population-level effects across task conditions and participants using a GLM/LME framework. 3) Hemodynamic variability: capturing the heterogeneity of the hemodynamic responses across different brain regions using basis functions and versatile modeling approaches. 4) Multiplicity challenge: Addressing the "multiple comparisons" problem inherent in massive univariate analysis through spatial regularization.

Introduction to P300 Brain-Computer Interface Data Analysis Challenges

Jane Huggins (University of Michigan)

Brain-computer interfaces (BCI) based on brain responses to flashing keys in an on-screen keyboard provide an intriguing access method for communication or computer access for people with disabilities. This BCI design (called a P300 BCI) overcomes the low signal-to-noise ratio of the brain signals by combining information from repeated presentations of stimuli. Signal variability caused by disability or changes in concentration can further complicate development of BCI algorithms. Specific challenges include 1) calibrating the BCI with a minimum amount of data to accommodate users with short attention spans, 2) differentiating between periods when the BCI user wants the BCI to make selections and when they want it to wait, 3) managing variance in signal latency as a user's attention fluctuates, and 4) managing variations in signal quality. The BCI design will be explained, data characteristics illustrated, and examples of BCI algorithms provided.

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DAY 2: June 2, 2026

7:30–8:00 AM

Coffee

8:00–9:20 AM

Track A · S12

Methods for Statistical Imaging Data with Applications to Physical Sciences

Organizer: Yang Chen · Chair: Yang Chen

David van Dyk (Imperial College London)

Identification of high-energy astrophysical point sources via hierarchical Bayesian nonparametric clustering

Light from distant astrophysical objects carries information about their origins and the physical mechanisms that power them. The study of astrophysical images is complicated because light from multiple localized sources is mixed and the sources are set against a spatially varying background. A general algorithm for robust source identification within these images remains an open question in astrophysics. This paper focuses on high-energy light (such as X-rays and gamma-rays), for which observatories detect individual photons, measuring their incoming direction, arrival time, and energy. We propose DeCaRD, a Bayesian source extraction method that involves deconvolution, clustering, and region detection. It uses spatial and energy information to identify point sources by deconvolving them from the spatially varying background, to estimate their number, and to probabilistically cluster photons into the identified sources. Because the data are pixilated, we detect regions made up of sets of pixels that contain one or more point sources. Point sources are modeled with a Dirichlet process mixture, while the background is simultaneously reconstructed via a flexible Bayesian nonparametric model based on B-splines. DeCaRD is validated with a suite of simulations and illustrated with an application to a complex region of the gamma-ray sky observed by the space-based Fermi Large Area Telescope.

Hengrui Luo (Rice University)

Bayesian Partition-Based Scalar-on-Image Regression with Group Spike-and-Slab Priors

We propose BPSiR-GSS: Bayesian Partition-based Scalar-on-Image Regression with Group Spike-and-Slab priors, a novel framework for regression models with image predictors designed to adaptively identify spatially contiguous regions associated with clinical outcomes. Unlike existing approaches that rely on an external fixed partition or a chosen atlas, our approach recursively partitions the spatial domain itself, producing an interpretable representation of region-specific effects. We employ a discrete group spike-and-slab prior on the partition nodes, which enables multi-level sparsity and region-level voxel selection. Theoretical considerations align the model with emerging posterior contraction results for discrete spike-and-slab priors in spatial settings. For posterior inference, we develop a trans-dimensional MCMC algorithm that employs a marginalization over the regression

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coefficients, resulting in an efficient updating scheme over partitions and sparsity patterns. Extensive simulation studies demonstrate that BPSiR-GSS achieves superior prediction accuracy and variable selection performance compared to state-of-the-art spatial Gaussian processes and low-rank tensor regression models. We apply the method to structural MRI data to predict brain age, successfully identifying localized patterns of neural degeneration that provide clean, interpretable maps of aging-related regions of interest. This is a joint work with Marina Vannucci at Rice University and Suprateek Kundu at M.D. Anderson Cancer Center.

Jeffrey Regier (University of Michigan)

Expectation-Maximization with Dual-Stream Flow Matching for Cataloging Astronomical Images

Upcoming astronomical surveys will produce petabytes of high-resolution images of the night sky, providing information about billions of stars and galaxies. Detecting and characterizing the astronomical objects in these images is a fundamental task in astronomy---and a challenging one, as most objects are faint and many visually overlap with other objects. In this talk, I present an expectation-maximization algorithm for probabilistic cataloging. In the E-step, neural posterior estimation (NPE) approximates the posterior distribution of catalogs under the current model using a continuous normalizing flow (CNF). This CNF is trained with a novel flow-matching loss function to jointly model both discrete latent variables (is there a galaxy here?) and continuous latent variables (if so, what are its properties?). In the M-step, the evidence lower bound (ELBO) is optimized to refine population-level priors for stars and galaxies. I demonstrate the fidelity and efficiency of this approach by processing terabytes of Dark Energy Survey images and tens of gigabytes of LSST DP1 coadds on a single Nvidia RTX 5090 GPU. Finally, I discuss extending this approach to infer luminosity functions, mass-richness relations, and cosmological parameters directly from survey images.

Rachel Mandelbaum (CMU)

Precise estimation of weak gravitational lensing distortions from images

Measurements of weak gravitational lensing, which causes a subtle but coherent distortion in galaxy shapes due to deflection of light by mass, require care in image processing. Numerous other effects, such as the blurring of the images due to the atmosphere and telescope optics, can cause distortions in galaxy shapes that are comparable in magnitude to the distortions caused by lensing. In this talk I describe the current state of the art in measurements of weak lensing from galaxy images in wide-field surveys, along with challenges that must be overcome in order to make the most of the latest generation of sky surveys starting in the near future and learn about fundamental physics from weak lensing with imaging surveys.

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Track B · S13

From Connections to Understanding: Brain Network Models in Neuroscience

Organizer: Heather Shappell · Chair: Heather Shappell

Yi Zhao (Indiana University School of Medicine)

Estimation of Heterogeneous Causal Mediation Effects

Mediation analysis is widely used to elucidate the mechanisms through which an exposure influences an outcome of interest. However, standard causal mediation methods may yield inconsistent conclusions, potentially due to unmodeled heterogeneity in mediation effects across individuals. To address this challenge, we propose a new framework that incorporates covariate-treatment and mediator-treatment interactions within a linear structural equation modeling system. Causal assumptions are discussed and heterogeneous natural direct and indirect effects are parameterized as functions of patient characteristics. A modified covariate approach is introduced to relax hierarchical constraints in models with interactions, and generalized lasso regularization is employed to achieve parsimony in high-dimensional settings. We establish asymptotic properties of the proposed estimators and demonstrate their finite-sample performance through extensive simulation studies. Application to data from the Alzheimer's Disease Neuroimaging Initiative reveals substantial heterogeneity in the mediation effect of cerebrospinal fluid volume change on the association between APOE- ϵ 4 carrier status and cognitive decline, identifying a subgroup of individuals who may be particularly susceptible to cognitive impairment through this disease pathway.

Yize Zhao (Yale School of Public Health)

Bayesian graph-informed modeling for Tau-connectivity interaction

Converging evidence indicates a complex interaction between Tau-protein propagation and alterations in brain connectivity. Aimed at characterizing Tau protein spread along functional networks in the early course of Alzheimer's disease, we develop a Bayesian graphical model to jointly model tau propagation, functional connectivity structure, and subgroup heterogeneity using cross-sectional data. By integrating graph-constrained infection dynamics with connectivity patterns, the model infers plausible propagation pathways and subgroup-specific infection sequences. We show interesting results on heterogeneous Tau accumulation patterns by applying the method to the A4 study, with the findings further extended to the ADNI study.

Mohsen Bahrami (Wake Forest University School of Medicine)

Brain Networks: Multivariate Statistical Tools to Study Structure, Function, and Dynamics

The availability of large-scale neuroimaging datasets and powerful computational resources has created an unprecedented opportunity to explore the structure, function, and dynamics of the brain as what it truly is—a complex system. However, conventional statistical models have inherent limitations that restrict the ability of clinicians and scientists to fully leverage the

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wealth of information contained in the neuroimaging data. To address these challenges, we have developed novel multivariate modeling frameworks and accompanying software tools, built on mixed-effects regression backbones, that enable relating phenotypic characteristics (e.g., age, disease phenotype, etc.) to whole-brain networks and specific subnetworks such as the default mode network (DMN) and drawing principled statistical inference. These modeling tools overcome critical limitations of existing methods and have successfully been applied across a range of studies examining the effects of various phenotypes including aging, weight loss, alcohol drinking, pesticide exposure, HIV and Marijuana use, and cerebral microbleeds on brain networks.

Noah Gade (Wake Forest University)

Segmentation in Dynamic Neuroimaging via Random Featurizations

The integration of statistical network analysis with neuroimaging has catalyzed a paradigm shift in our understanding of brain function. Inferring these directed interactions, however, is particularly challenging when functional connections are not static, but nonstationary. This talk bridges dynamic network analysis for neuroimaging with temporal segmentation. We introduce a flexible and adaptive random featurization framework to discern abrupt and gradual structural changes in the connectivity network, avoiding the traditional computational burden of deep learning methods. Post-segmentation, network inference is simplified, allowing for recovery of directed information flow within each quasi-stable regime.

Track C • S14

Advanced Statistical Modeling and Machine Learning for Brain Imaging Data

Organizer: Qiong Wu • Chair: Qiong Wu

Aiying Zhang (University of Virginia)

ConnectMVR: A Supervised Brain Connectivity Analysis Framework for Predicting Behavioral Outcomes

Brain connectivity—the complex network of neural pathways enabling communication among brain regions—provides a powerful framework for understanding individual differences in cognition, emotion, and behavior. This project introduces ConnectMVR, a penalized multivariate matrix-variate regression framework designed for supervised brain connectivity analysis and multimodal neuroimaging integration. ConnectMVR identifies statistically reliable brain-behavior associations across multiple connectivity matrices at both the regional and connection levels, while accounting for the high dimensionality and structured dependencies inherent in connectome data. Using data from the Human Connectome Project in Development (HCP-D), we will apply ConnectMVR to integrate intrinsic functional connectivity from resting-state fMRI and structural connectivity from diffusion MRI,

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linking these measures to behavioral assessments. This framework offers a rigorous, interpretable, and generalizable approach for uncovering multiscale connectivity patterns underlying behavioral variability in youth.

Shuo Chen (University of Maryland Baltimore)

Trustworthy ML/AI for Aging Clocks: Preventing Systematic Prediction Bias in Biological Age Estimation

Machine learning (ML)-based aging clocks have become widely used as objective measures of an individual's underlying physiological and molecular age based on omics data, as compared to chronological age. In this work, we show from a computational perspective how the inherent systematic prediction bias of commonly used ML regression models induces bias in aging clocks and thereby distorts and in some cases even reverses, downstream biomedical conclusions, for example, estimated associations between biological age and risk factors. We first formally quantify ML-induced bias in aging clocks and analytically derive how this bias propagates to downstream association estimates. Simulation studies and two case studies, including brain age and epigenetic clocks, are highly consistent with these theoretical results. Together, these findings highlight the importance of quantitatively evaluating the systematic bias of aging clocks to support valid biomedical inference. We further introduce an unbiased ML/AI regression framework as a principled solution to this problem.

Yuan Wang (University of South Carolina)

Topological Inference and Clustering of Functional Brain Networks

Modeling individual variability in brain functions is essential for understanding neurological disorders such as post-stroke aphasia. We propose a unified topological framework to characterize this heterogeneity and its impact on both behavioral deficits and treatment response. Our approach begins with a novel method for lesion-symptom mapping (LSM) that leverages persistent homology to capture higher-order topological features in functional brain networks. By moving beyond region-wise or edge-level comparisons, this method identifies mesoscale structures—such as cycles—that reflect distributed lesion effects. Statistical inference is performed via heat kernel representation of persistence diagrams and a permutation-based testing procedure. We also extend this framework to investigate treatment response by introducing a topological clustering method based on heat-kernel representation of persistence diagrams, enabling integration of Euclidean covariates and automated selection of the number of clusters. Together, these methods provide a cohesive topological perspective on brain-behavior relationships, offering multiscale, data-driven tools for analyzing functional brain networks in neuroimaging studies.

Ruiwen Zhou (Washington University in St. Louis)

A Temporal Graph Neural Network for Time-to-Alzheimer's Disease Prediction Using Longitudinal Neuroimaging Data

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by gradual structural brain changes that precede clinical symptoms. The Alzheimer's Disease Neuroimaging Initiative (ADNI) provides longitudinal structural MRI (sMRI)

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data—cortical thickness, surface area, and volumetric measures across brain regions—that serve as important biomarkers for tracking early AD progression. However, predicting time-to-AD conversion from these data is challenging due to inter-subject variability, complex spatial dependencies among brain regions, and temporal heterogeneity across visits. Traditional and deep learning classifiers often overlook censoring and fail to jointly model spatiotemporal patterns of neurodegeneration. We propose a temporal graph neural network (GNN) framework for time-to-Alzheimer's prediction that integrates anatomical structure with longitudinal trajectories. Our approach combines landmark analysis to handle censoring with a GNN that encodes structural relationships among brain regions, along with LSTM/Transformer-based temporal modules that capture both local dynamics and long-range progression. These hybrid architectures jointly model spatial organization and evolving sMRI patterns, producing individualized and clinically meaningful time-to-event risk predictions. The proposed framework offers a flexible and powerful tool for early detection of AD and may improve the identification of high-risk individuals in longitudinal neuroimaging studies.

9:30–10:50 AM

Track A • S15

Statistical Method for Interpreting Astronomical Images

Organizer: Jeffrey Regier • Chair: Jeffrey Regier

Kevin McKinnon (University of Toronto)

Measuring precise motions of the faintest stars by combining Hubble, James Webb, and Gaia

The Gaia satellite has provided first-ever positions and motions for billions of stars over the last decade. However, observing constraints mean that Gaia's measurements get worse for faint stars and truncate past a certain brightness. These faint stars are particularly interesting for many open astrophysical questions, such as the nature of dark matter and the precise formation history of the Milky Way. Using my BP3M tool (Bayesian Positions, Parallaxes, and Proper Motions), I am able to combine Hubble images with Gaia data to measure order-of-magnitude improved stellar motions, especially for the faintest stars. In this talk, I will explain BP3M's robust statistical techniques, highlight ongoing applications with different astronomical questions, and present work expanding to James Webb data. I will also present in-progress applications of modern computer vision techniques, such as Neural Radiance Fields (NeRFs), to improve calibration of the complex distortions present in telescope images. Finally, I will discuss the future of BP3M-like methods and ML-enhanced calibration for the next generation of telescopes and surveys (e.g., Euclid, Roman, Rubin/LSST).

Tamás Budavári (Johns Hopkins University)

Towards Cleaner and Deeper Sky Reconstructions from Astronomy Exposures: ImageMM and Beyond

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The quality of ground-based photometric observations is limited by the atmosphere in several ways. Our goal is to optimally combine multiple exposures to obtain the best possible angular resolution while also maintaining the high signal to noise of traditional coadds. ImageMM tackles the ill-defined inversion problem using the majorize-minimize algorithm to find feasible solutions. I will discuss the latest results as well as our ongoing efforts and plans to further improve the templates and detection images for source detection for current surveys, such as the Rubin Observatory's Legacy Survey of Space and Time, and for future experiments.

Andrew Saydjari (Princeton University)

Separating Stars and the Diffuse Sky: Leveraging Thermal Light Echoes for 3D Tomography

Images in astronomy can be represented as a combination of several components: point sources (e.g., stars), extended compact sources (e.g., distant galaxies), and diffuse emission. Separating images into these individual components is required for precision measurements of the colors of stars/galaxies and to unveil the structure of gas/dust in the interstellar medium. In this talk, I will describe recent progress in Bayesian component separation techniques for astronomical imaging, using data-driven priors on each component. I will use as a case study a recent imaging program from JWST that reveals the turbulent substructure of dust in the interstellar medium using thermal light echoes. From the "clean" diffuse component of the images, we can learn about the hydrodynamic properties of the interstellar medium. By leveraging a time series for tomography, we use the diffuse components and Gaussian process methods to achieve the highest Cartesian resolution 3D dust map to date. I will conclude by highlighting the scalability of this class of component separation methods and their application to some of the largest surveys in astronomy, including SPHEREx.

Yang Chen (University of Michigan)

Probabilistic Tensor Prediction for High-Dimensional Solar Imaging Data

The increasing availability of high-resolution solar imaging and heliophysical time series motivates new statistical methods for probabilistic space weather forecasting. This talk presents two related modeling developments: (i) a Tensor Gaussian Process with Contraction for scalar-on-tensor regression, which reduces multi-channel images to an interpretable latent tensor using tensor contraction with anisotropic total-variation regularization, then applies a multilinear-kernel Tensor-GP for probabilistic prediction; and (ii) a model for forecasting matrix-valued time series on spatial grids using past images and auxiliary vector time series, combining row- and column-wise autoregressive structure with an RKHS-based tensor coefficient, estimated via penalized maximum likelihood and alternating minimization. Applications to solar flare prediction and to forecasting global geomagnetic indices are used to evaluate performance. The talk concludes with directions for operational solar eruption forecasting and monitoring of corresponding terrestrial impacts.

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Track B · S16

Statistical and AI Methods for Heterogeneous and Irregular Imaging Data

Organizer: Chunming Zhang · Chair: Chunming Zhang

Haochang Shou (University of Pennsylvania)

Adaptive Shrinkage Estimation for Personalized Deep Kernel Regression in Modeling Brain Trajectories

Longitudinal biomedical studies monitor individuals over time to capture dynamics in brain development, disease progression, and treatment effects. However, estimating trajectories of brain biomarkers is challenging due to biological variability, inconsistencies in measurement protocols (e.g., differences in MRI scanners) as well as scarcity and irregularity in longitudinal measurements.

Herein, we introduce a novel personalized deep kernel regression framework for forecasting brain biomarkers, with application to regional volumetric measurements. Our approach integrates two key components: a population model that captures brain trajectories from a large and diverse cohort, and a subject-specific model that captures individual trajectories. To optimally combine these, we propose Adaptive Shrinkage Estimation, which effectively balances population and subject-specific models. We assess our model's performance through predictive accuracy metrics, uncertainty quantification, and validation against external clinical studies. Benchmarking against state-of-the-art statistical and machine learning models—including linear mixed effects models, generalized additive models, and deep learning methods—demonstrates the superior predictive performance of our approach. Additionally, we apply our method to predict trajectories of composite neuroimaging biomarkers, which highlights the versatility of our approach in modeling the progression of longitudinal neuroimaging biomarkers. Furthermore, validation on three external neuroimaging studies confirms the robustness of our method across different clinical contexts.

Liyue Shen (University of Michigan)

Constrained and Controllable Diffusion Models for Computational Imaging

Diffusion models have recently emerged as powerful generative priors for solving inverse problems. However, training diffusion models is data-intensive and computationally demanding, which restricts their applicability for high-dimensional and high-resolution data such as medical imaging. In this talk, I will introduce our recent works on how to improve the efficiency (data, time, and memory efficiency) of diffusion-based generative models for solving general inverse problems through constrained posterior sampling. Particularly, I will introduce two promising solutions we propose to enable learning diffusion priors for solving high-dimensional imaging problems through latent diffusion and patch diffusion models. The results are demonstrated in solving various inverse problems for both natural and medical images including 3D medical image reconstruction, showing the effectiveness of our proposed methods in both model efficiency and model performance. If time permits, I will also introduce our recent works to

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enhance the generalization and controllability of diffusion sampling. Our recent findings in the initial noise space reveal new opportunities for uncertainty quantification, strengthening weak prior, and enabling fine-grained control in image restoration and reconstruction tasks. This research opens the door to leverage diffusion-based generative models in tackling complex real-world data for addressing crucial inverse problems in many biomedical and scientific applications.

Christopher Geoga (University of Wisconsin–Madison)

Sharpening and accelerating Fourier methods for irregularly sampled measurements

Function recovery from irregular Fourier samples is a ubiquitous problem in a variety of fields. When measurements in k-space are not given on a lattice, however, a variety of artifacts and numerical issues can become problematic sources of signal pollution. In this work, I will discuss a recent advance in the design of weighting schemes to significantly mitigate these problematic artifacts that borrows from the theory of numerical quadrature, which can be adapted to a wide variety of sampling settings and geometries while retaining attractive algorithmic runtime costs. Particularly in the case of densely sampled and highly non-gridded data, this approach provides significant improvements over current popular weighting methods such as density compensation factors. After a discussion of the method and a few theoretical observations, demonstrations in exploratory analysis of spatio-temporal data and statistical inverse problems will be discussed.

John Bodenschatz (Marquette University)

Mathematically Accurate Maximum Likelihood Estimation Detects Activation in fMRI Phase Data from Local Magnetic Field Changes

In functional magnetic resonance imaging (fMRI), it is important to observe the functioning brain as fast as possible and at as high of a spatial resolution as possible. Increased spatial and temporal speed results in voxels with increased noise relative to signal and contrast. There is much evidence to suggest that there is important biological information contained within the phase component of the fMRI signal which is directly linked to local changes in magnetization. When the signal-to-noise ratio within a voxel is low, as when there is ultra-high resolution, the marginal statistical distribution of the phase is non-standard and difficult to work with. This non-standard marginal phase distribution at high signal-to-noise ratios is Normally distributed, but at low signal-to-noise ratios needs to be utilized for accurate modeling. In this work, phase-only activation will be computed directly from Lathi's mathematically correct non-Normal distribution, yielding additional physiological information beyond what is typically observed.

Track C • S17

Achieving Reliable Inference and Prediction from Noisy Imaging Data through Structure Learning

Organizer: Shuo Chen • Chair: Sharmistha Guha

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Fangyi Wang (Ohio State University)

Joint Registration and Conformal Prediction for Partially Observed Functional Data

Predicting missing segments in partially observed functions is challenging due to infinite-dimensionality, complex dependence within and across observations, and irregular noise. These challenges are further exacerbated by the existence of two distinct sources of variation in functional data, termed amplitude (variation along the y-axis) and phase (variation along the x-axis). While registration can disentangle them from complete functional data, the process is more difficult for partial observations. Thus, existing methods for functional data prediction often treat phase variation as negligible. Furthermore, they typically require precise model specifications and/or rely on computationally intensive tools, such as bootstrapping, to construct prediction intervals. We propose a unified registration and prediction approach for partially observed functions using conformal prediction. Our method integrates registration and prediction in one algorithm while ensuring exchangeability through carefully constructed predictor-response pairs. Using a neighborhood smoothing algorithm, the framework produces pointwise prediction bands with finite-sample marginal coverage guarantees under weak assumptions. The method is easy to implement, computationally efficient, and permits simple parallelization. Numerical studies and real-world data examples demonstrate the effectiveness and practical utility of our method.

Megan Jones (Vanderbilt University)

Estimation of Maximum Achievable Predictive Accuracy for Machine Learning Brain-Phenotype Associations

Machine learning is used in neuroscience to examine brain-phenotype associations and facilitate individual prediction from high-dimensional brain imaging. For continuous phenotypes, Pearson's correlation between the observed and predicted phenotype is used to quantify model accuracy in testing data. However, recent research suggests millions of samples may be needed to reliably estimate the maximum achievable predictive accuracy (MAPA). We formally define the MAPA and show that Pearson's estimator is biased for this quantity and its confidence intervals fail to capture the target. We develop a semiparametric (double machine learning) one-step estimator that more accurately estimates the MAPA and yields valid confidence intervals across flexible machine learning settings. Analyzing data from the Reproducible Brain Charts dataset, we show that this estimator has smaller bias when estimating brain-phenotype associations of neuroimaging data with age and psychopathology phenotypes. We further extend by improving the estimation of area under the curve (AUC) for MAPA for binary phenotypes.

Yezhi Pan (University of Maryland)

Network-Guided Multivariate Regression for High-dimensional Imaging Outcomes

Mass-univariate analysis remains the workhorse of brain-wide association studies, where tens of thousands of imaging-derived phenotypes are modeled independently. Yet neuroimaging outcomes demonstrate strong, structured correlation patterns driven by

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latent brain network organization, and ignoring this dependence structure reduces statistical efficiency and limits replicability. We present a network-guided multivariate regression framework that integrates covariance network structure into multivariate regression analysis to explicitly account for inter-correlations among neuroimaging measures. This approach provides an alternative to conventional mass-univariate modeling with dependence adjustment, analogous to the role of mixed-effects models for repeated measures in lower-dimensional settings. Through extensive simulations and an application to imaging-derived phenotypes from the ABCD study, we show that network-guided modeling substantially improves inferential accuracy and cross-study replicability.

Jiaao Yu (University of Maryland)

Structured multi-modal imaging interaction discovery with a triple-graph mode

TBD

11:00–11:15 AM

Break

11:15 AM–12:15 PM

Keynote 2: Prof. Nicole Lazar (Penn State) (Michigan League Ballroom)

On the Replicability of Functional Brain Networks

Problems of replicability plague many areas of science, including neuroimaging. Prior research has pointed to the large number of preprocessing pathways available for functional magnetic resonance imaging (fMRI) data and implications for downstream analyses and replicability. One important aspect of replicability in the context of functional brain networks is their consistency across the various preprocessing decisions that a researcher might make. The lack of standardized procedures serves as a potential source of heterogeneity in the resultant networks. In this talk, I explore the effects of certain preprocessing choices on seven commonly studied functional networks derived from resting state fMRI data, using the Autism Brain Imaging Data Exchange (ABIDE) as a test case. While some choices have a large impact on the estimated strengths of network connections, others play a lesser role. I will also discuss ways to assess differences in network structure, or lack of consistency, stemming from these researcher degrees of freedom.

Joint work with Kaitlyn Fales, Xurui Zhi, and Hyebin Song.

12:15–1:30 PM

Lunch

1:30–2:50 PM

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Track A · S18

Recent Advances in Complex Modeling of Brain Imaging and Networks

Organizer: Xi Luo · Chair: Jian Kang

Panpan Zhang (Vanderbilt University Medical Center)

Group-Structured Sparse Precision Estimation for Interpretable Functional Brain Networks

Functional brain networks estimated from fMRI data are increasingly used in cognitive analyses. However, most statistical methods impose sparsity at the level of individual connections. We propose a group-structured sparse precision matrix framework for estimating functional connectivity that aligns statistical regularization with scientifically meaningful network organization. The proposed approach performs model selection over groups of connections, yielding interpretable network estimates. The framework is formulated through a general penalized-likelihood approach that accommodates multilevel group structures and unifies several existing sparse precision estimators as special cases. We investigate identifiability conditions, scalable optimization algorithms for high-dimensional settings, and stability tools for assessing the reliability of selected network groups. Simulation studies demonstrate improved recovery and stability of group-level structure relative to existing competing methods. An application to the Tennessee Alzheimer's Project shows how group-structured sparsity produces brain networks that are more interpretable and more directly related to cognitive outcomes. All reproducible simulation code and user-friendly utility functions are provided in an open-source R package.

Marina Vannucci (Rice University)

Bayesian Brain Network Mediation Analysis of Longitudinal Alzheimer's Disease Data

Causal mediation analysis provides critical insights into how exposures influence outcomes through intermediate variables, or mediators. In this study, we examine mediation effects of complex-structured data represented as whole-brain connectivity networks derived from resting state fMRI (rs-fMRI) data on Alzheimer's Disease (AD) subjects and healthy controls. Our modeling approach incorporates low-dimensional latent scales representations of the high-dimensional connectivity matrices, while preserving node-level characteristics and facilitating the identification of key mediating brain regions. It also accommodates flexible non-linear relationships between the mediators and the outcomes of interest. Using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI-1), we perform cross-sectional and longitudinal analyses to investigate the effects of several factors known to affect cognitive decline. Results confirm the important role of amyloid-beta as a risk factor for cognitive impairment. They also confirm the key mediation role played by the default mode network, reinforcing its central role in AD. The longitudinal analysis additionally identifies a distinct subset of brain network nodes, beyond those found in the cross-sectional analysis, that uniquely drive baseline and/or time-varying indirect causal effects, particularly in the visual network.

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Suprateek Kundu (The University of Texas MD Anderson Cancer Center)

Bayesian Semi-parametric Tensor-on-Tensor Regression for Forecasting Alzheimer's Disease Progression

There is an increasing interest in tensor-on-tensor (TOT) modeling literature with a particular focus on image-on-image regression (IIR). However, key limitations exist, including restrictive linearity assumptions, the absence of local spatial context, and limited capacity to capture heterogeneity across samples. We propose a novel Bayesian TOT modeling framework to address these gaps designed for longitudinal IIR settings with a focus on image forecasting that is underdeveloped in literature. The TOT model integrates voxel- and subject-specific local varying coefficient (VC) terms, that capture non-linear heterogeneous longitudinal patterns and is modeled using Gaussian processes (GP), with low-rank global tensor coefficients encoding spatial information. To further capture local spatial dependencies, we incorporate a patch-to-voxel mapping, enabling each output voxel to depend on its spatial neighborhood in the input image.

We develop an efficient Markov chain Monte Carlo (MCMC) algorithm that exploits the low-rank tensor structure and an embarrassingly parallel update scheme for voxel-specific GP components, enabling scalable inference for high-dimensional images with tens of thousands of voxels. Extensive simulations demonstrate substantial gains over existing methods in terms of image prediction accuracy, while also exhibiting computational scalability and MCMC convergence.

Applied to longitudinal T1-weighted MRI data from the Alzheimer's Disease Neuroimaging Initiative, the proposed approach accurately forecasts future cortical thickness changes based on imaging data from past visits. The resulting forecasted images enable downstream estimation of accelerated brain aging at future time horizons, providing a biologically meaningful marker of future disease progression that is estimated by accounting for concurrent neurodegenerative changes.

Yu-Ping Wang (Tulane University)

Nonlinear mixed models for brain network analysis based on Multi-Paradigm fMRIs

Functional magnetic resonance imaging (fMRI) data are inherently complex, characterized by high dimensionality, intricate inter-regional dependencies, and substantial individual variability across experimental paradigms. Traditional linear mixed models (LMMs) often fail to adequately capture nonlinear relationships inherent in neuroimaging data. To address these limitations, we introduce nonlinear mixed model (NMM), an extension of the LMM framework that integrates neural networks to flexibly model complex fixed-effect relationships while preserving the random effects structure to account for individual differences. We applied the NMM to Philadelphia Neurodevelopmental Cohort (PNC) across emotion, n-back, and resting-state paradigms, which achieved superior model fit relative to classical LMMs. This framework offers a statistically rigorous and practically explainable approach for modeling large-scale functional connectivity from modest covariates while explicitly separating population-level effects from stable individual variability in functional brain organization.

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Track B · S19

Next-Generation Statistical and Inferential Methods for Neuroimaging and Multimodal Data Integration

Organizer: Haochang Shou · Chair: Haochang Shou

Zhirui Li/Bingxin Zhao (Department of Statistics and Data Science, University of Pennsylvania)

Agentic AI in imaging genetics

AI agents can serve as virtual research collaborators in imaging genetics, accelerating discovery and translation by orchestrating complex, end-to-end workflows that span genomic data processing, imaging phenotype engineering, association and prediction modeling, replication, and interpretation. Realizing this potential requires robust domain-specific datasets, standardized pipelines, and scalable computational infrastructure, as well as interdisciplinary collaboration to establish community standards that ensure transparency, reproducibility, and responsible deployment. In this talk, we will present an agentic framework for imaging genetics, highlight practical use cases in complex disease and aging-related brain phenotypes, and discuss how human-in-the-loop, source-traceable evidence synthesis can improve real-world utility of data resources and AI tools.

Jeffery Morris (Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania)

Scalable distributional regression

Many modern instruments, such as wearables, imaging devices, and geospatial sensors, produce rich subject-level data streams containing thousands to millions of measurements. Reducing these data to simple summaries (e.g., means) can obscure meaningful structure. We introduce a general distributional regression framework for distribution-on-scalar or distribution-on-function regression, enabling characterization of how subject-specific distributions vary across conditions and covariates. Our modeling strategy applies a series of transformations to each subject's empirical distribution function, producing a compact, low-dimensional embedding that is near-lossless under the quantile-domain Wasserstein metric. These embeddings preserve essential distributional information, accommodate measurement discreteness arising from instrument precision, satisfy Gaussian assumptions required for regression in the embedding space, and maintain monotonicity in the quantile domain. We further describe efforts to construct generative models that produce valid subject-level distributions reflecting the salient structure of the underlying population and examine the statistical properties of inference derived from these models, and discuss inferential strategies.

Quy Cao (Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania)

Spatial Bootstrap for Voxel-Wise Neuroimaging Analysis

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Imaging data are often summarized by a single value per region of interest, a practice that can reduce statistical power and obscure important spatial heterogeneity. Voxel-wise analyses better capture the complexity of imaging data, but accurate estimation of standard errors in this setting is challenging due to spatial correlation among voxels. We propose a spatial bootstrap framework for voxel-level inference that accounts for spatial dependence while preserving statistical power. The method is demonstrated through an application examining the association between Quantitative Susceptibility Mapping and Myelin Water Fraction, illustrating its ability to provide reliable standard error estimates and robust statistical inference.

Noah Hillman (Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania)

Testing for Network Specificity in Brain-Behavior Associations Using Ordinal Dominance Curves

Interpreting brain-behavior relationships through the lens of anatomical parcellations or functional networks is commonplace in human brain mapping. However, statistical approaches for testing whether brain-behavior associations are stronger (i.e., enriched) within a region of interest remain underdeveloped. Here, we propose a permutation-based approach for network enrichment testing using ordinal dominance curves (NETDOM). In simulation studies, we demonstrate that NETDOM properly controls the type I error rate—unlike other prominent methods—while exhibiting increased statistical power when brain-behavior associations are elevated in a subset of in-network locations. Using data from two large-scale neurodevelopmental cohorts, we illustrate that NETDOM effectively detects enriched associations between structural and functional brain measures and neurocognitive performance.

Track C • S20

AI for Medical Image Analysis II

Organizer: Kayvan Najarian • Chair: Shuo Chen

Morgen Henry, Emily Wittrup & Kayvan Najarian (University of Michigan, Ann Arbor)

A Four-Stage Statistical Pipeline for Medical Image Super-Resolution via Generative Adversarial Learning

The fidelity of medical diagnostic images, such as those acquired via ultrasound, directly influences the validity of downstream statistical inference and clinical decision-making. Low-resolution (LR) images impose information loss that classical deterministic upscaling methods, such as bicubic interpolation, cannot recover. These traditional methods operate effectively as low-pass filters, attenuating high-frequency structural content and introducing systematic bias in downstream classification and inference tasks. Alternatively, we propose a four-stage pipeline that leverages an Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) as a generative engine for super-resolution, integrated with a rigorous statistical verification process to ensure that the resulting high-resolution (HR) images represent a measurable improvement in structural detail over the original LR data. The pipeline begins with a preprocessing stage that standardizes image intensities and corrects for acquisition-related variation across

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heterogeneous inputs, ensuring the model learns biologically meaningful structure rather than scanner-specific artifacts. The generative core employs a relativistic discriminator paired with a perceptual loss functional, framing super-resolution as estimation on the HR image manifold rather than pointwise mean prediction. This overcomes the well-known regression-to-the-mean phenomenon induced by mean squared error-based objectives, which produces posterior mean estimates that systematically suppress high-frequency variation. A subsequent standardization stage applies pixel-level normalization and stochastic augmentation to stabilize empirical feature distributions prior to downstream analysis. The pipeline concludes with a formal statistical verification step, auditing pixel intensity distributions and screening for non-biological artifacts, providing a principled quality-control procedure analogous to residual diagnostics in classical regression. This work frames medical image super-resolution explicitly as a statistical estimation problem, and offers the imaging community a reproducible, auditable pipeline for enhancing LR datasets in ways that demonstrably improve sensitivity and specificity of diagnostic models.

Khashayar Namdar et al. (University of Toronto)

RadiomiXAI: adjusting radiomics feature importance via Bayesian Network-guided explainability

Radiomics features often show high intercorrelation due to latent factors like tumor volume, distorting feature importance in ML models. We present RadiomiXAI, a model-agnostic explainability framework using Bayesian networks to adjust importance scores by modeling feature dependencies. Using 476,424 radiomics feature vectors extracted from OpenRadiomics via PyRadiomics, we computed Pearson correlations with tumor volume and identified 74 features with correlation coefficients >0.8 , suggesting strong dependency. Multiple normalization techniques were evaluated for variance inflation factor (VIF) reduction. A LightGBM classifier was trained on BraTS 2020 T1 Contrast Enhanced MRI data using a 60/20/20 train/validation/test split. Feature importances were estimated using RadiomiXAI, a perturbation-based method that replaces each feature with sampled values from its range and quantifies the mean absolute change in predicted probabilities. Uncertainty intervals for each importance score were then derived using the BN model: incoming dependencies reduce the score (lower bound), and outgoing dependencies inflate it (upper bound), proportional to mutual information values on the BN edges. Normalization techniques failed to adequately decorrelate most features from volume. RadiomiXAI identified an outgoing dependency-adjusted upper bound of 0.2059 for volume's base importance score of 0.0087, revealing its potential downstream overrepresentation. RadiomiXAI reveals how latent common-cause variables such as tumor volume can inflate feature importance scores in radiomics pipelines. Conventional normalization fails to resolve these dependencies. BN-guided uncertainty bounds provide a novel explainability framework to adjust feature interpretations, aiding robust biomarker identification.

Nooshin Maghsoodi (Queen's University)

Deep learning with ECG data in the ICU: From modelling to actionable AI

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Deep learning with ECG data in the ICU is not only about building accurate models, but about creating AI tools that can truly help doctors make better decisions. Atrial fibrillation (AFib) is one of the most common and serious heart rhythm problems in critically ill patients, and can increase the risk of complications and death. However, ICU ECG signals are noisy and full of artifacts, and constant alarms make it hard to reliably detect these brief AFib events. In this project, we follow the evolution of AI methods for AFib detection in this challenging environment. We begin with classical machine learning based on carefully designed ECG and heart-rate-variability features, then move to deep neural networks that learn directly from raw signals, first trained on large public datasets and then adapted to ICU data using transfer learning and weak labels. We next explore large ECG foundation models pretrained on huge datasets and fine-tune them for ICU use. Finally, we develop our own large model trained from scratch using physiologically guided self-supervised contrastive learning, allowing us to fully exploit the vast amount of unlabeled ICU ECG data. Together, this work shows how ICU ECG analysis can progress from traditional modeling to powerful, robust, and clinically actionable deep learning systems.

Amoon Jamzad & Parvin Mousavi (Queen's University)

Spatial Prompting for Representation and Exploration of High-Dimensional Data

Spatial high-dimensional data is now routine in computer vision. Modern vision foundation models produce dense embeddings that capture rich semantic structure across an image. Similar spatial high-dimensional measurements also arise in mass spectrometry imaging, where each location contains a molecular signature. These representations are valuable for discovery and downstream decision-making. However, they are difficult to interpret directly. A standard approach is global dimensionality reduction to form a low-dimensional view. In practice, global projections can wash out subtle variation and local heterogeneity. We propose an interactive spatial prompting method, with supporting software, that uses user-provided spatial cues to guide dimensionality reduction. This produces higher-contrast views that better support exploratory analysis.

3:00–4:20 PM

Track A • S21

Recent advancements in statistical methods for MRI

Organizer: Benjamin Risk • Chair: Benjamin Risk

Seonjoo Lee (Columbia University)

Model Selection for Exposure-Mediator Interaction for high-dimensional imaging mediators

In mediation analysis, the exposure often influences the mediating effect, i.e., there is an interaction between exposure and mediator on the dependent variable. When the mediator is high-dimensional, it is necessary to identify non-zero mediators (M) and exposure-by-mediator (X-by-M) interactions, research is scarce in preserving the underlying hierarchical structure between the main

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effects and the interactions. To fill the knowledge gap, we develop the XMInt procedure to select M and X-by-M interactions in the high-dimensional mediators setting while preserving the hierarchical structure. Our proposed method employs a sequential regularization-based forward-selection approach to identify the mediators and their hierarchically preserved interaction with exposure. Our numerical experiments showed promising selection results. Further, we applied our method to ADNI morphological data and examined the role of cortical thickness and subcortical volumes on the effect of amyloid-beta accumulation on cognitive performance, which could be helpful in understanding the brain compensation mechanism.

Chao Huang (University of Georgia)

Dynamic Mediation Analysis

Mediation analysis decomposes the effect of an explanatory variable on an outcome into direct and indirect components through mediators. Conventional mediation models assume constant effects, which can be overly restrictive when causal relationships vary with covariates. We propose a dynamic high-dimensional mediation framework based on varying coefficient models that allows the effects of the explanatory variable on both the mediator and the outcome to change smoothly with covariates. Estimation is carried out using B-spline approximations, with an Adaptive Lasso penalty to enable variable selection and control model complexity in high-dimensional settings. We establish convergence rates and asymptotic distributions for the proposed estimators. For inference, we develop an F-test for direct effects and a partially penalized Wald test for indirect effects, both of which are shown to have chi-square limiting distributions under the null, with noncentral chi-square behavior under local alternatives for the direct effect test. Simulation studies demonstrate the validity and robustness of the proposed methods. An application to Alzheimer's Disease Neuroimaging Initiative MRI data identifies region-of-interest-based volumetric mediators and reveals the dynamic influence of age on Alzheimer's disease. Overall, the proposed method provides a flexible tool for causal inference in high-dimensional environments.

Kaidi Kang (Wake Forest University School of Medicine)

Characterizing Longitudinal Trajectories of Structural MRI-derived Biomarkers Toward Alzheimer's Disease Progression Using a Double anchoring events-based Sigmoidal Mixed Model

Accurately modeling longitudinal trajectories of structural MRI (sMRI)-derived biomarkers along Alzheimer's disease (AD) progression is critical for understanding disease dynamics, yet remains challenging because available datasets typically capture only fragmentary segments of an individual's decades-long disease course. As a result, standard longitudinal analyses are often limited to subsets of participants with observed clinical conversion, and therefore, suffer from reduced statistical power and selection bias. To address these challenges, we previously proposed a double anchoring events-based sigmoidal mixed model (DSMM) and applied it to characterize cognitive decline trajectories using time to incident AD diagnosis as a clinically meaningful

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time scale, while uniquely enabling inclusion of participants without an observed incident AD diagnosis. In this work, we extend the DSMM framework to the characterization of longitudinal sMRI-derived biomarkers in participants with AD. Using harmonized multi-cohort data from the Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium (ADSP-PHC), the proposed approach aligns heterogeneous imaging trajectories relative to AD onset and integrates information from participants without observed AD conversion. This unified modeling strategy reduces selection bias, provides excellent clinical interpretability, and enables estimation and comparison of the population-level temporal ordering of changes in multimodal measures along the AD progression continuum.

Jun Young Park (University of Toronto)

Statistical considerations in intermodal coupling analysis

Within-individual coupling between measures of brain structure and function evolves across development and may underlie differential risk for neuropsychiatric disorders. Despite growing interest in the developmental trajectories of structure–function relationships, rigorous methods to quantify and test individual differences in coupling remain limited. In this talk, we outline a framework for defining intermodal coupling and parameterizing individual differences, and show how several existing approaches can be interpreted as statistical inference via cluster enhancement under specific assumptions. We then discuss methods for estimating the genetic heritability of intermodal coupling and for conducting valid statistical inference, highlighting key methodological challenges. We illustrate these approaches using data from the Philadelphia Neurodevelopmental Cohort (PNC) and the Human Connectome Project (HCP). Overall, our findings raise a cautionary note regarding intermodal coupling analyses, particularly with respect to proper control of false positives. This is joint work with Ruyi Pan and Ruilin Bai.

Track B · S22

Statistical Network Analysis for Neuroimaging

Organizer: Dan Kessler · Chair: Dan Kessler

Liza Levina (University of Michigan; Department of Statistics)

Comparing groups of networks

Network-structured data arise in many imaging applications, for instance, in fMRI neuroimaging. Their statistical analysis requires methods for modeling multiple networks, which have so far mostly focused on estimating the shared structure only. Two-sample tests have also been developed, for some version of the hypothesis of two samples of networks being globally indistinguishable. However, scientifically relevant hypotheses rarely take this form: in neuroimaging, it is rarely of interest to compare whole brains of patients and healthy controls, and the focus is more often on a particular brain region. Beyond comparisons, it is also of interest to

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estimate structures that correspond to a specific subgroup of networks, for example, patients with a certain trait or diagnosis. One could always do that using just the relevant subgroup of interest – either the one brain region or the one subgroup of patients – but using all available samples allows us to better estimate structures that are shared by all, increasing statistical efficiency. This talk presents two methods that take advantage of this general idea: mesoscale testing, which conducts a formal hypothesis test on a subset of edges while leveraging the rest of the network to increase power, and group MultiNeSS, which separates structures shared by all from those specific to subgroups or individuals. Both methods rely on latent space models and leverage the assumption of an underlying low-rank structure, which has been observed widely in practice. Based on joint work with Peter MacDonald, Alexander Kagan, and Ji Zhu.

Jaroslav Harezlak (Indiana University; School of Public Health)

Simultaneous Nuclear Norm and L_1 Regularization for Binary-outcome-on-Matrix Regression with Applications to Brain Imaging
We introduce a novel convex optimization framework for logistic scalar-on-matrix regression which incorporates nuclear and L_1 norm penalties to simultaneously enforce low-rank and sparse structures in the estimated coefficient matrix. Proposed method enables interpretable modeling of high-dimensional matrix-valued predictors in presence of binary responses. We derive a custom algorithm based on the Alternating Direction Method of Multipliers (ADMM) to efficiently solve the resulting convex optimization problem and establish the theoretical properties of the obtained solution. Numerical experiments demonstrate the effectiveness of our method in recovering meaningful predictive patterns. Finally, we apply our method to the brain imaging data to identify structures in functional brain connectivity matrices that are characteristic of subjects with a family history of alcohol use disorders (AUDs).

Keith Levin (University of Wisconsin–Madison; Department of Statistics)

Estimating Multiple Weighted Networks with Node-Sparse Differences and Shared Low-Rank Structure

We study the problem of modeling multiple symmetric, weighted networks defined on a common set of nodes, where networks are observed from different groups or conditions. We propose a model in which each network is expressed as the sum of a low-rank structure shared across conditions and a node-sparse matrix that captures the differences between conditions. This formulation is motivated by settings such as those seen in connectomics, where most nodes share a global connectivity structure while a few exhibit condition-specific changes in behavior. We develop a multi-stage estimation procedure that combines a spectral initialization step, semidefinite programming for support recovery, and a debiased refinement step for low-rank estimation. We establish minimax-optimal guarantees for recovering the shared low-rank component under the row-wise ℓ_2 norm and elementwise ℓ_∞ norm, as well as for detecting node-level perturbations under various signal-to-noise regimes. We

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demonstrate that the availability of multiple networks can significantly enhance estimation accuracy compared to single-network settings.

Braden Scherting (Duke University; Department of Statistical Science)

Quasi-Poisson multiplex ANOVA with applications to brain connectors

We seek to characterize differences in brain network structure between old and young individuals (across sex and lineage) and identify composite subnetworks. The motivating data come from a large cohort of mice and comprise counts of white matter fiber connections between pairs of predefined regions of interest. The data include approximately equal numbers of zeros, counts between 1 and 10, and counts exceeding 1000. We introduce quasi-Poisson multiplex ANOVA to robustly model these counts and enable inference on network-valued group contrasts. Our approach leverages factor regression featuring a multiplicative specification of the familiar log-linear ANOVA. An additional dispersion term is introduced at the latent level to induce overdispersion in the observed counts while preserving computational tractability, yielding a quasi-posterior for Poisson factorization. Identifiability and computational considerations, comparisons to Poisson and Negative Binomial multiplex ANOVA, initial results, and directions of further study are presented.

Track C · S23

Recent developments on neuroimage analyses for use in digital twins

Organizer: Ansu Chatterjee · Chair: Emily Hector

Siddhartha Nandy (University of Maryland Baltimore County)

Spatio-temporal modeling of neuroimages using Covariate Assisted Principal Regression

In this talk, we explore brain connectivity through covariate-assisted principal component regression. The nature of the connectivity explores the impact of external covariates on classical principal component analysis. The computational bottleneck arises from the high number of voxels common in fMRI. We implement the methodology on a dataset obtained from the OpenfMRI database, comprising 16 subjects of different genders and ages. To circumvent the computational challenge of the voxel-wise BOLD signals, we parceled the brain region into atlases using geometric parcellation, functional parcellation, and a hybrid of the two approaches. We present the functional network for the first two principal components and the roles of gender and age as covariates in identifying the principal components. Our study shows that gender and its interaction with age exhibited a significant impact on the first principal component for both functionally informed parcellation. For the geometric parcellation, despite the lack of demographic modulation, the latent components reveal a spatially distributed network structure; however they do not map onto the specific functional network that vary by age and gender in this cohort.

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Ansu Chatterjee (University of Maryland Baltimore County)

Rapid change detection in streaming neuroimage data

Early detection of neurodegenerative diseases is critical in ensuring proper treatment and care of patients, and in containment of the disease. Owing to the plasticity of the brain, neuroimage data are often governed by individualized parameters that make such rapid detections challenging. We propose techniques that combine statistical and artificial intelligence methods that leverage knowledge and information from multiple individuals, known neuroscience and biological sciences, and modern computational techniques for rapidly identifying changes in neuroimages from any single individual. Thus, our framework combined the power of using a cohort, yet potentially delivers inferences applicable in a personalized healthcare setting. Algorithmic details and numerical results will be presented, and some theoretical findings outlined.

Sanjay Chaudhuri (University of Nebraska, Lincoln)

An empirical likelihood-based method for functional neuroimaging

The empirical likelihood (EL) method has long been established as a powerful tool in many areas of statistics. This class of semi-parametric procedures enables parametric inference through constrained nonparametric estimation of underlying distribution functions. In this work, we present an EL-based framework for detecting task-related activation in functional magnetic resonance imaging (fMRI) data. The proposed method evaluates the significance of task-related covariates at each voxel within the standard regression framework for fMRI analysis. The null distribution of the EL-based test statistic is estimated empirically using voxels outside the brain region, where no activation is expected. Active voxels are subsequently identified by comparing voxel-wise test statistics with this estimated null distribution, followed by false discovery rate control. The methodology is illustrated using real fMRI data involving visual stimuli consisting of famous faces, unfamiliar faces, and scrambled images. The resulting activation maps demonstrate that the proposed approach successfully identifies biologically meaningful brain regions associated with visual processing tasks.

4:30–6:30 PM

[SMI-2026 Posters and Mixer Event](#)

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DAY 3: June 3, 2026

7:30–8:00 AM

Coffee

8:00–9:20 AM

Track A · S24

Network methods in neuroimaging

Organizer: Liza Levina · Chair: Liza Levina

Tingting Zhang (University of Pittsburgh)

Age-Related Changes in Structural Networks Across Large Neuroimaging Studies

Understanding how structural connectivity (SC) evolves across the human lifespan is critical for elucidating brain development, aging, and disease. While functional connectivity has been extensively studied, characterization of age-related changes in SC remains limited and inconsistent across studies, largely due to methodological heterogeneity and the inherent sparsity of structural brain networks. In this study, we provide a systematic and large-scale investigation of lifespan changes in SC between all pairs of brain regions using diffusion MRI data aggregated from multiple cohorts spanning development to aging. Our approach enables a comprehensive characterization of age- and sex-related changes in structural brain networks at the connection level. By leveraging large, heterogeneous datasets and principled statistical modeling, this work provides a more reliable and detailed understanding of lifespan trajectories of SC and offers a general framework for analyzing sparse network data in neuroimaging.

Russel T. Shinohara (University of Pennsylvania)

Multi-Site Harmonization Methods for Network-based Analyses

Magnetic resonance imaging (MRI) studies often leverage artificial intelligence (AI) and machine learning in large multi-center designs. To address the issue of systematic biases between scanners, an increasingly commonly used approach is ComBat which was first proposed in the genomics literature. Our group has found these methods helpful but sometimes insufficient in multivariate scenarios such as those involving network analysis, including when conducting personalized network modeling. To address this, a broader framework has been developed for multi-scanner harmonization, allowing for flexible, non-linear, and multivariate modelling in various imaging science settings.

Daniel Kessler (University of North Carolina, Chapel Hill)

Predicting Responses from Weighted Networks with Node Covariates in an Application to Neuroimaging

We consider the setting where many networks are observed on a common node set, and each observation comprises edge weights of a network, covariates observed at each node, and an overall response. The goal is to use the edge weights and node covariates to

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predict the response while identifying an interpretable set of predictive features. Our motivating application is neuroimaging, where edge weights encode functional connectivity measured between brain regions, node covariates encode task activations at each brain region, and the response is disease status or score on a behavioral task. We propose an approach that constructs feature groups based on assumed community structure (naturally occurring in neuroimaging applications). We propose two feature grouping schemes that incorporate both edge weights and node covariates, and we derive algorithms for optimization using an overlapping group LASSO penalty. Empirical results on synthetic data show that our method, relative to competing approaches, has similar or improved prediction error along with superior support recovery, enabling a more interpretable and potentially more accurate understanding of the underlying process. We also apply the method to neuroimaging data from the Human Connectome Project. Our approach is widely applicable in neuroimaging where interpretability is highly desired.

Alexander Huth (University of California, Berkeley)

Mapping and Decoding Language Representations from Human Cortex

Is it possible to read the content of human thought out using recordings of brain activity? We use non-invasive functional MRI and machine learning methods based on large language models to investigate the relationship between brain activity and the content of thought. These methods reveal complex spatial and temporal patterns of activity that relate to specific semantic categories. We show that this information can be read out as language, even when the stimulus evoking it is from another modality. We also use self-supervised speech models to show that fine temporal information can be deduced even from slow non-invasive recordings. These results point to a future of neuroscience that strongly integrates modern neural network models.

Track B • S25

AI for Medical Image Analysis III

Organizer: Kayvan Najarian • Chair: Ivo Dinov

Pascal Tyrrell Khashayar Namdar, et al. (University of Toronto)

Restoring Radiomic Feature Interpretability After Principal Component Analysis in Prostate MRI Cancer Risk Models

To enhance prostate cancer risk stratification through explainable artificial intelligence (XAI) by developing a pipeline that integrates principal component analysis (PCA) for dimensionality reduction and restores interpretability in radiomics-based classification using multiparametric MRI. We used the 2021 PI-CAI dataset from RadMLBench, comprising 3,045 radiomics features extracted from T2-weighted, apparent diffusion coefficient (ADC) maps, and diffusion-weighted imaging (DWI) MRI sequences of 969 patients. The task involved binary classification of csPCa (Gleason >3+3, n=637) vs. non-csPCa (n=332). We applied 10-fold nested cross-validation using LightGBM and compared baseline performance with a dimensionality-reduced variant utilizing PCA preserving 95%

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of variance. The baseline 10-fold AUC without dimensionality reduction was 0.701. Incorporating PCA improved performance, achieving an AUC of 0.753. On a representative 60/20/20 split, the test set yielded an AUC of 0.735, accuracy of 0.727, and F1-score of 0.819. To overcome PCA's limitation in post-hoc interpretability, a feed-forward neural network with a single hidden layer (256 units) was trained to reconstruct the full radiomics space. Feature importance scores from LightGBM were projected through this mapping and transformed into the original feature domain using the stored eigenvector matrix. Post-hoc interpretability analysis identified top contributing features in the original radiomics space. This study addresses the challenge of interpretability in high-dimensional radiomics datasets by reconciling dimensionality reduction with explainable machine learning, supporting transparent prostate cancer risk modeling.

Daljit Takher, Kayvan Najarian & Omer Berenfeld(University of Michigan, Ann Arbor)

Physics-Learning of Rotor Dynamics: A Reaction-Diffusion Video Prediction Framework for Cardiac Fibrillation

Ventricular and atrial fibrillation are major causes of morbidity and mortality with enormous clinical-societal burden. Recent experimental and computational research studies suggest that fibrillation can be sustained by functional re-entrant patterns of excitation, also termed rotors, but their dynamics remain unclear. In fundamental terms, the heart is considered an excitable medium whose spatio-temporal patterns of excitation are governed by a reaction-diffusion process. Thus, we propose a physics-informed video prediction framework for modeling rotor-driven excitation dynamics using simulated voltage-map sequences generated from FitzHugh-Nagumo reaction-diffusion systems. The model embeds physics-inspired inductive bias through a latent space representation, and then performs reaction-diffusion-based temporal updates, and spatial coupling via a fixed Laplacian operator, enabling interpretable forecasting of excitation dynamics from spatiotemporal voltage maps.

Rosa Fallahpour, Khashayar Namdar & Pascal. N. Tyrrell(University of Toronto)

A Spatial Entropy Framework for Characterizing Radiomic Heterogeneity: Application to Healthy and Pediatric Tumor Brain MRI

Quantifying spatial heterogeneity in medical imaging is important for understanding structural variability and improving imaging-derived biomarkers, which can be beneficial for machine learning-based modeling. In this work, we introduced a Shannon entropy-based methodology designed to quantify spatial heterogeneity in brain MRI using radiomics-derived traits. In this approach, each brain image was partitioned into regions, and radiomics features were extracted from each region using a standardized radiomics pipeline. Continuous feature values were then transformed into categorical traits, allowing the construction of trait distributions across spatial regions. The proposed methodology summarizes the overall spatial heterogeneity of MRI images by aggregating entropy-based heterogeneity measures derived from these trait distributions. This formulation provides an interpretable metric that captures the diversity of radiomic patterns across brain regions. The proposed framework offers a systematic and reproducible approach for translating radiomics features into quantitative measures of spatial complexity, enabling comparative heterogeneity

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analysis across medical imaging datasets and supporting future studies in developing heterogeneity-aware machine learning models trained by MRI images. The framework was evaluated on two brain MRI datasets: the IXI healthy adult dataset ($n = 270$) and the BraTS-PED pediatric brain tumor dataset ($n = 251$). Using first-order radiomic entropy as an example feature and dividing each brain into four spatial quadrants, the proposed method produced heterogeneity average index (HTA) of 0.820 for the IXI dataset and 0.884 for the BraTS-PED dataset. Individual quadrant-level indices (HTI) exhibited substantial variance (0.618 to 0.995), reflecting the tool's sensitivity to localized structural diversity. These results indicate that the framework provides a standardized scale for comparing spatial complexity across diverse cohorts, offering a necessary precursor for integrating heterogeneity-aware features into predictive machine learning models.

9:30–10:50 AM

Student Paper Session

(Michigan League Ballroom)

11:00–11:15 AM

Break

11:15 AM–12:15 PM

Keynote 3: Prof. Douglas Noll (University of Michigan) (Michigan League Ballroom)

Spatiotemporal signal modeling in dynamic MRI

Spatiotemporal signal models in dynamic imaging have continued to grow in importance. We explore the role of these models in dynamic magnetic resonance imaging (MRI) for both denoising and image reconstruction from sparse samples. Specifically, we examine partially separable functions (low-rank representations) as well as other signal representations derived from data and signal simulations. We also describe some applications, such as functional neuroimaging using MRI (functional MRI), which are particularly challenging due to the low contrast signals.

12:15–1:30 PM

Conclusion/Adjourn